Rooftop Solar or a New Kitchen: Do Rooftop Solar Systems Capitalize into Home Values?

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Abstract

This paper estimates the impact of residential rooftop solar systems on housing resale prices in Delaware and examines heterogeneity across system attributes. We use single-family home transactions and solar installation data to estimate solar price premiums using both hedonic and repeat-sales models. We find that homes sold with solar systems receive an average price premium of 7.6% ($22,379) when recent home renovations are not controlled for and 5.6% ($16,613) after controlling for them. These premiums exceed the average net installation cost of a residential solar system in Delaware ($14,781, after the federal Investment Tax Credit), implying that homeowners recover approximately 112%–125% of their solar investment upon resale. Capitalization is strongest within the first three years after installation and declines thereafter. Solar value does not disappear with age; rather, it shifts from capitalized resale value to realized electricity bill savings. We further develop counterfactual policy simulations that combine resale premiums with electricity bill savings under alternative federal, state, and utility incentive policies. The simulations show that federal Investment Tax Credits and state incentives significantly accelerate investment recovery, whereas removing these incentives delays cost recovery by approximately ten years. We also find that premiums are higher for owned than leased systems and are larger in metropolitan than non-metropolitan markets. The findings provide new evidence from a low-penetration solar market and offer insights for homeowners, policymakers, and solar market stakeholders designing effective solar policies.

Keywords: Hedonic Analysis, Rooftop solar system, Residential housing market, Home renovations
JEL Codes: Q42, R31, Q48, Q51

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1 Introduction

Over the past decade, the cost of residential rooftop solar systems in the United States (U.S.) has declined substantially, driven by technological improvements, economies of scale, supportive federal and state policies, and solar renewable energy certificates. As a result, there has been a notable rise in solar system adoption, with more households choosing to install these clean energy solutions. Incorporating these systems into our energy infrastructure offers a practical solution to the long-term climate impacts of non-renewable energy sources, as they provide sustainable energy and reduce energy costs.

Despite this growth, solar power remains a relatively small part of the U.S. electricity market. According to the U.S. Energy Information Administration (EIA), solar accounted for approximately 5.1% of total U.S. electricity generation in 2024, while utility-scale electricity generation was about 4.30 trillion kilowatt-hours (kWh), and about 0.08 trillion kWh generated from small-scale solar systems (EIA, 2025b). Earlier federal policy ambitions have been far more expansive: the U.S. Department of Energy (DOE) has projected that solar could supply roughly 40% of national electricity generation by 2035 (Ardani et al., 2021). Achieving this goal will require substantial increases in residential adoption, particularly in states where adoption remains low. However, since 2025, the current federal energy policy has shifted toward non-renewable energy sources and has no specific goal for the solar energy share.

Residential rooftop solar adoption has expanded rapidly in a limited number of U.S. states, yet adoption remains uneven across households and regions. Prior research identifies a range of barriers to adoption, including high upfront installation costs, homeownership constraints, limited information about solar systems and related policies, roof suitability, insufficient sunlight, uncertainty about long-term savings, and concerns about payback periods, and difficulties finding trustworthy installation companies (Zhang et al., 2023; Rai et al., 2016; Palm, 2018; Gillingham and Tsvetanov, 2019; Agathokleous and Kalogirou, 2020; Shakeel et al., 2023). Another notable concern among prospective adopters is whether solar investments are capitalized into home values at the time of resale – will homeowners get their money back on their system if they choose to or have to sell their home? Relatively little attention has been given to how expectations about future home resale value influence solar investment decisions.

Rooftop solar systems differ from many other residential investments because they generate electricity bill savings over the lifespan of the system, 25-30 years (Walker and McDevitt, 2025). However, homeowners do not necessarily remain in the same property long enough to realize these benefits. Rresidential mobility is common in the U.S., 11.8% of the population (39.7 million people) changed residences in 2024 (U.S. Census Bureau, 2025). This implies that many homeowners may sell their homes well before a solar investment reaches its full payback period. For these households, the financial return to solar depends not only on future electricity savings but also on whether the remaining economic value of solar is capitalized into the home's resale price. Uncertainty about resale value can discourage homeowners from adopting solar even when long-run energy savings are substantial. Therefore, it is important to understand whether and to what extent solar systems are reflected in housing prices, as this is critical for household decision-making and policy design.

Delaware provides a particularly informative setting to explore these questions. Although the state offers many of the same federal and state incentives available elsewhere, residential rooftop solar adoption remains limited. Delaware is a small state with 471,318 housing units (U.S. Census 2023), and only about 8,736 homes (1.85%) have installed solar systems by 2025. This adoption rate is considerably lower than in leading solar states despite similar policy frameworks (See Figure A1). The low-penetration solar market provides an important test of whether solar capitalization exists and extends beyond the high-penetration solar market. One hypothesis is that we would expect weaker or no capitalization in low-penetration solar regions because prospective buyers are less familiar with solar, face uncertainty regarding its costs and benefits, worry about maintenance and system performance, and observe relatively few comparable transactions involving solar homes. If so, homeowners may be less able to recover their investment through resale, reinforcing slow adoption. On the other hand, finding capitalization despite a low-penetration market would suggest that barriers other than weak economic returns play a more important role in explaining slow adoption.

Moreover, the eastern U.S. remains underrepresented in the solar hedonic literature, which has focused disproportionately on western and southwestern housing markets. Delaware's relatively small size, combined with the availability of high-quality administrative data on solar installations, property transactions, and renovation permits, makes it a valuable case study for understanding solar capitalization in an understudied and low-penetration region.

Installing a residential solar system not only contributes to environmental sustainability but also brings about a diverse range of financial returns, both direct and indirect. By installing solar panels, households can generate their own energy, which in turn reduces their reliance on purchasing electricity from the grid and ultimately leads to significant savings on their monthly electricity bills. The availability of state and federal incentives, such as investment tax credits (ITC), solar renewable energy certificates (SRECs), state rebates, low- to moderate-income solar programs, and renewable portfolio standards (See Section 5.4 Table 4 and Appendix subsection A.4), expands the range of financial returns available to households. Additionally, net metering allows solar households in specific locations not only to offset their energy consumption but also sell surplus electricity back to their utility company. Beyond these energy-related benefits, solar systems may also yield non-energy returns by increasing the home's market value. From an economic perspective, rooftop solar can be viewed as a durable housing attribute, akin to other structural or energy-efficiency improvements, that may be capitalized into sale prices through buyers' willingness to pay.

The relationship between housing attributes and property values has been extensively studied in the real estate economics literature. Hedonic pricing models consistently show that standard characteristics, structural, locational, neighborhood amenities, school quality, environmental, and informational attributes of housing are capitalized into home values (Rosen, 1974; Cheshire and Sheppard, 1998; Haag et al., 2000; Bond et al., 2002; Pryce and Oates, 2008; Kuminoff Nicolai V. and Timmins, 2013; Nowak and Smith, 2017). The literature also documents that renovations and capital improvements yield more value at resale (Harding et al., 2007), and that environmental and energy attributes can be capitalized into home and building values (Eichholtz et al., 2010; Brounen and Kok, 2011; Hyland et al., 2013). Within this framework, rooftop solar installations can be interpreted as an additional housing attribute or a quality improvement that may enhance property value.

Why is it important to explore the effect of solar on house sale prices? The installation of solar panels is an expensive and time-consuming process; however, consumers don't know how long they will remain in the same house, which discourages them from adopting solar. For example, if you install the solar system on your home, it costs you $X now, and after a few years, you want to sell your house with a solar system. Will you lose or gain money? Are you able to recover your solar investment in the housing market? These are possible questions that might be on consumers' minds and make them reluctant to adopt solar power; therefore, investigating the effect of solar installation on house value is important. The solar housing market has been examined in previous studies in the U.S., the United Kingdom (UK), and Australia. A seminal solar housing market study in two California cities shows that price premiums of around 3-4% for solar homes (Dastrup et al., 2012). Hoen et al. (2013) found similar results in more cities across California. Similarly, other studies find comparable or larger price premiums in different U.S. states, in Hawai'i (Wee, 2016), in Arizona (Qiu et al., 2017), and in Massachusetts-Connecticut (Gillingham and Watten, 2024). Finally, these results have been replicated in Australia (Ma et al., 2016; Lan et al., 2020), and in the UK (Asproudis et al., 2024). Overall, this literature offers strong evidence that rooftop solar systems are capitalized into housing prices at the time of sale.

However, most prior studies on this question did not directly control for recent home renovations, except for (Gillingham and Watten, 2024; Asproudis et al., 2024). This omission may bias the estimates of capitalization upwards or downwards. Anecdotally, properties are often renovated or improved shortly before and after sale, including roof upgrades during solar installation, electrical work, major additions, utility renovations, and other renovations that independently increase property values. In the absence of detailed renovation data, it is difficult to disentangle the price effects of solar systems from those of concurrent home improvements.

We used detailed home-renovation permits and historical property data to address this gap in the literature on the solar housing market. We construct a novel property-level dataset by combining four datasets: solar installation records obtained from a state agency, historical residential property characteristics, property transaction data, and detailed renovation permit records from Cotality (formerly CoreLogic), a leading provider of U.S. real estate, property transaction, assessment, and building permit data. The inclusion of renovation records is particularly important because it allows us to control for unobserved confounders. To our knowledge, we are the first to use historical records of residential property characteristics in a solar hedonic context, which contain important information about major additions. The historical and permit data allow us to observe the timing and type of home improvements before sale and to explicitly control for recent renovations across multiple categories, including structural work, major additions, and energy upgrades.

The main objective of this paper is to estimate the effect of a solar system on the house price premium in Delaware, a relatively low-penetration solar market. We extend our investigation by demonstrating heterogeneity across solar system attributes and spatial contexts. In particular, we examine whether price premiums vary by system age, system size, ownership structure (owned versus leased systems), and metropolitan versus non-metropolitan location. Finally, we assess how federal and state solar policies affect investment recovery by combining resale premiums with other solar benefits under alternative policy scenarios. To address identification concerns, we employ both hedonic price models and repeat-sales specifications with census block group, property, and year-quarter fixed effects, thereby controlling for time-invariant unobserved spatial heterogeneity and broader market trends. We are optimistic that our estimated results get closer to showing the causal effect of solar on housing values. The homes sold with solar systems have an average price premium of 5.6% ($16,613), after controlling for recent home renovations. This is higher than the average installation cost of a solar system in Delaware, which is $14,781, after the federal ITC. This suggests that solar is fully capitalized at the time of resale and recovers 112% of solar cost. However, we observe that solar systems were fully capitalized when homes were sold within the first three years of installation, but that capitalization rates decline afterward. The solar value does not disappear with age; rather, it shifts from capitalized resale value to realized electricity bill savings. The counterfactual policy simulations show that federal ITC and state incentives substantially accelerate solar investment recovery, whereas removing these incentives delays cost recovery by about ten years. We also found that homes with owned solar systems had a higher price premium than homes with leased solar systems. Additionally, capitalization effects were greater in metropolitan areas than in non-metropolitan regions.

The remainder of this paper is organized as follows. In section 2, we review the broad literature on hedonic studies of housing characteristics, renovation, energy attributes, and prior contributions to the solar housing market. In section 3, we discuss the four datasets used in this analysis. In section 4, we discuss the empirical strategy, which includes the hedonic and repeat-sales approaches. We present the results and discussion in section 5, and finally, the conclusions in section 6.

2 Literature Review

2.1 Housing characteristics and the hedonic pricing framework

The determinants of housing prices have long been studied in urban and real estate economics, with evidence consistently showing that market values reflect a combination of macroeconomic conditions, structural attributes, and locational amenities. Macroeconomic fundamentals, such as interest rates, unemployment, inflation, Gross Domestic Product (GDP) growth, mortgage rates, equity market performance, and public spending, play an important role in shaping housing demand and price dynamics (Vaidynathan et al., 2023). At the property level, key structural characteristics such as floor area, number of bedrooms and bathrooms, land area, property age, and amenities including garages, pools, and air conditioning are consistently identified as the most important determinants of housing values across markets and time periods (Sirmans et al., 2006). In addition to structural features, location-specific factors such as neighborhood quality, access to high-quality schools, transportation infrastructure (i.e., access to major highways), retail services, public transit, recreational amenities, employment hubs, and neighborhood environmental quality significantly influence property values (Abelson, 1979; Clapp and Giaccotto, 1998; Crompton, 2001; Netusil, 2005; Clapp et al., 2008).

Housing is treated as a differentiated good in the hedonic pricing framework to study the relationship between its characteristics and price. According to Rosen's (1974) established theory, housing prices are interpreted as households' willingness to pay for structural, locational, and neighborhood characteristics. This framework provides the primary empirical foundation for analyzing housing markets and is widely used to estimate the marginal value of individual characteristics.

Some research indicates that housing quality is not fully captured by the above-mentioned home characteristics. This information embedded in real estate listings, seller descriptions, and other textual data can convey important signals about unobserved property condition and seller motivation, influencing both listing prices and transaction outcomes (Haag et al., 2000; Bond et al., 2002; Pryce and Oates, 2008; Kuminoff Nicolai V. and Timmins, 2013; Nowak and Smith, 2017). These findings highlight a persistent challenge in hedonic modeling: unobserved or imperfectly measured quality differences can bias estimates of attribute capitalization when correlated with the variables of interest.

Older homes are usually sold at lower prices, however, if they have undergone major renovations, including kitchens, bathrooms, and major additions, this can increase home resale value (Miller et al., 2018). We would consider a rooftop solar installation to represent one of many categories of possible major additions to a home. Anecdotally, homeowners sometimes cluster renovation activities in time, including getting a new roof shortly before installing solar, implying that renovation activity may not only be an unobserved covariate, but may, in fact, be correlated with solar installation in time. According to a recent study by Heinrich et al. 2020, roof age is a key determinant of rooftop solar suitability, as solar systems are long-lived assets that typically operate for 25-30 years and require a roof with sufficient remaining useful life. Simultaneous roof replacement and solar installation would reduce homeowners' acquisition costs for rooftop solar by around 20% (Heinrich et al., 2020). Similarly, homeowners and solar installers are frequently advised to replace an aging roof before or during solar installation to avoid the future costs of removing and reinstalling solar when roof replacement becomes necessary, as both have similar lifespans (DOE, 2021; Mooney et al., 2025). Because roof replacement itself can increase property values, solar installation may coincide with substantial capital investments in the home. Therefore, a true estimate of solar capitalization requires accounting for all home characteristics and renovation activities.

2.2 Home renovations and unobserved quality change

A related strand of the literature focuses on housing quality change over time, particularly the role of renovations, remodeling, and structural improvements. Unlike many housing attributes that depreciate gradually, renovations can generate discrete and sometimes substantial changes in housing quality and value. Kitchen and bathroom remodels, major additions, roof replacements, and structural upgrades have all been shown to increase resale values, although the magnitude of returns varies by renovation type and market conditions.

Failure to observe renovation activity can lead to biased estimates in both hedonic models and repeat-sales analyses. Bogin and Doerner (2019) demonstrates that unobserved renovations can distort measured house price appreciation and bias housing price indices when quality changes are incorrectly attributed to market trends rather than property-specific investment. Similarly, standard repeat-sales indices assume constant property quality between transactions; however, unobserved changes such as renovations and redevelopment can violate this assumption and bias price change estimates (Clapp and Giaccotto, 1998; Nowak and Smith, 2019).

Despite the recognized importance of renovations, empirical studies often lack detailed, transaction-level data on home improvements. As a result, researchers typically rely on proxy controls, such as property age or broad time trends, which may inadequately capture recent investments in housing quality. This limitation is particularly salient in settings where the adoption of a new technology or amenity is correlated with renovation behavior. In such cases, estimated capitalization effects may reflect a combination of the technology itself and contemporaneous, unidentified improvements.

2.3 Capitalization of energy attributes

The economic foundation for studying the effect of rooftop solar systems on home prices follows the standard hedonic theory of housing markets, in which durable attributes of a home are capitalized into market prices through buyers' willingness to pay (Rosen, 1974; Freeman, 1979). The capitalization of rooftop solar systems can be understood through a simple present-value framework that reflects the stream of future electricity bill savings they generate. Let S_t denote the expected annual electricity bill savings produced by a solar system in year t. Assuming prospective buyers are discounting future benefits at rate r, the value of the solar system can be approximated by the discounted present value (PV) of expected future savings: $Premium_t = \sum_{s=t}^{T} \frac{E(S_s)}{(1+r)^s}$, where T represents the remaining useful life of the system. This implies that solar premiums should decline as systems age because buyers capitalize only the remaining future benefits rather than the original installation cost (See more discussion in section 5.3 and Figure 2).

This fundamental theoretical prediction aligns with broader literature demonstrating that energy-related housing attributes are capitalized into property values. Studies of green building certifications (Eichholtz et al., 2010; Brounen and Kok, 2011), and residential energy-efficiency ratings (Brounen and Kok, 2011; Hyland et al., 2013) show that energy-efficient or environmentally certified properties sell at a positive price premium relative to non-certified properties. Similarly, distributed renewable-energy installations, including solar hot-water systems, have been shown to yield positive capitalization effects (Qiu et al., 2017; Gillingham and Tsvetanov, 2019; Guo et al., 2024). Solar systems fit squarely within this theoretical and empirical framework because they provide measurable economic benefits, are highly visible to potential buyers, and are supported by federal, state, and utility-level policies that enhance their financial value.

2.4 Rooftop solar capitalization: evidence from the United States

The U.S. literature on solar capitalization began with regional studies and has subsequently expanded to multi-state analyses. A seminal study on solar capitalization by Dastrup et al. (2012) investigates the effect of solar system installations in two cities, San Diego and Sacramento, California, and finds that homes with solar systems sell at a 3% to 4% premium, using hedonic and repeat-sales methods. This indicates a $22,554 increase in solar home prices, representing full capitalization after considering subsidies. Importantly, they show that premiums are higher in neighborhoods with stronger liberal/environmental preferences, providing early evidence that the "green" amenity value is heterogeneous. They also find a larger capitalization of 7% for homes located on streets with few nearby solar installations, compared with 3% for a solar home on a "solar street"—streets where at least two adjacent homes have rooftop solar systems. Lastly, the larger effect of 6.2% is observed when controlling for a small number of building renovations (Dastrup et al., 2012).

A landmark contribution by Hoen et al. (2013) analyzes 1,894 California solar home sales and finds that, in general, solar home prices are 3.6% higher (of average size 3.20 kW) than non-solar home prices (yields a premium of $5.5/W or $17,000). Specifically, they find that the average solar premium for a new solar home is $2.3-2.6/W, whereas for an existing solar home it is $6-7.7/W, creating a larger disparity. Their follow-up multi-state study (Hoen et al., 2017), using 22,822 sales across eight states, reports premiums of approximately $3.5–$4/W or $15,000 for an average-sized 3.6kW, which are robust across markets and time periods. A notable insight from this work is that local electricity prices and policy incentives significantly affect capitalization. Their results consistently show that buyers internalize both cost savings and remaining value of the solar system (Hoen et al., 2013, 2017).

Another related U.S. case by Wee (2016), which studies O'ahu Hawai'i, where electricity prices are the highest in the country and there are the highest number of solar installations per capita nationwide. She finds average solar premiums of 5.4% (approximately $35,000), consistent with strong bill-savings incentives and grid-circuit constraints. These results demonstrate that policy context and electricity tariffs heavily influence capitalization. Similarly, Qiu et al. (2017) in a study of the Arizona solar housing market, employ hedonic and matching methods and find that the solar-home transaction price premium is a 17% ($28,000) increase at a medium sales price.

Gillingham and Watten (2024) re-examine how solar is capitalized into home prices in Massachusetts and Connecticut. They emphasize challenges related to unobserved renovations, selection into solar adoption, and the bundling of solar system installations with roof replacements, issues that many early studies could not fully address. They also develop a theory-driven model to understand solar capitalization and find full capitalization for an owned solar system and lower levels of capitalization for leased systems. Taken together, the U.S. findings are remarkably consistent in showing positive premiums of roughly 3–17%, though magnitudes vary across geographic regions.

2.5 Rooftop solar capitalization: evidence from Australia & UK markets

Outside the U.S., Australia has produced some of the most thorough evidence due to its high penetration of rooftop solar. Ma et al. (2016) were the first to analyze Western Australian data and found significant solar home price premiums of 2.3% to 3.2% ($16,124 to $22,240), suggesting full-capitalization, consistent with U.S. results. Lan et al. (2020) examine Southport in Queensland, Australia, and document that the solar premium declines from 4.3% to 2.4% as feed-in tariffs (FiTs) fall from 44 cents/kWh to 8 cents/kWh. Their net-benefit calculations indicate that, after FiT reductions, homeowners recoup a much smaller share of the installation cost. This case study provides clear evidence that policy generosity directly affects capitalization (Lan et al., 2020).

Similarly, Best et al. (2021) provides a large-scale national study and shows that solar premiums vary substantially by region, policy regime, and housing type. Beyond solar home sales markets, Best et al. (2023) analyze solar home rental markets and estimate that a 5% increase in weekly housing rents (Australian $19) suggests that landlords have benefited from investments in solar systems through higher rents. This indicates that capitalization varies sharply with tenure and incentive incidence.

The recent study by Asproudis et al. 2024 investigates the returns on solar panels in the UK residential housing market using a Machine Learning (ML) approach, and their findings suggest that the selling price increases by £14,062 to £16,368 for houses with a solar system (Asproudis et al., 2024).

2.6 Methodological approaches and limitations

Methodologically, the literature has evolved from simple cross-sectional hedonic models toward more sophisticated panel, repeat-sales, and quasi-experimental designs. Table A1 summarizes the existing literature on residential solar capitalization. Taken together, this literature suggests an opportunity to examine the effects of solar system age on capitalization and whether estimates are sensitive to detailed controls for home renovations. Additionally, there is a greater scope to investigate heterogeneity in solar ownership, which has been explored only by Begley and Hoen (2021) and Gillingham and Watten (2024). Existing studies also provide limited evidence on how changes in federal, state, FiT, and utility incentives affect the overall financial returns to solar systems. Lastly, the external validity of existing estimates remains uncertain because prior studies are heavily concentrated in high-adoption markets, particularly California and Australia. Therefore, investigating the Delaware solar housing market is important for understanding housing markets with lower solar penetration and test whether solar capitalization exists and extends beyond high-penetration solar regions. (see Figure A1).

3 Data Sources

We used three separate property datasets and one solar installation dataset to conduct the analysis reported here. Below are specifics of each:

3.1 Rooftop solar installation data

We obtained data on solar installations in Delaware (DE) for 2002-2025 from the Department of Natural Resources and Environmental Control (DNREC), Delaware. The dataset contains over 9,226 unique solar installations on both residential (8,792) and commercial properties (434). However, we restrict the sample to 8,792 residential solar installations. Some households (56) upgrade their solar system (increase capacity), therefore, we count these homes only once. After accounting for these upgrades, we end up with a total of 8,736 installations. This data includes information on system type (e.g., residential, commercial), system ownership status (e.g., owned or leased "Third Party Owner (TPO)"), detailed system information (e.g., solar capacity in kW, system installation calendar date), and financial information (e.g., renewable energy certificate factor, total installed costs). It also includes residential address (street, block, city, county, and zip code). This data indicates that about 25% household has leased/TPO solar systems and the remaining 75% have owned solar systems. Other important features we used in our analysis include solar system capacity; the average capacity is approximately 8.30 kW in this dataset. The average cost of a solar system is $34,467 before the government subsidy, not adjusted for inflation. However, inflation-adjusted by the 2010 CPI, the costs are $25,672 (all solar) and $21,115 (home sale with solar), which further decline with incentives to $17,970 (all solar) and $14,780 (home sale with solar), respectively (See Appendix Table A9).

3.2 Property sale data

We obtain Delaware property data from Cotality (CoreLogic), a private supplier of U.S. real estate data. Specifically, we obtain three different datasets that include historical property characteristics data, transaction data, and permit data. These are thoroughly cleaned separately and used to construct a unique transaction-level cross-section and repeat-sale data.

3.2.1 Historical property data

We use historical residential property characteristics data, which is important to control for property renovation. This dataset contains the characteristics of each property, including location, address, historical tax records, and property attributes, which have been updated in each tax year. We obtain fifteen waves of these datasets, differentiated by tax roll certificate date, and create panel data by appending them. All waves were appended for the years

2009-2024 into a single file, yielding a total of 7,164,244 observations.

A unique identification number assigned to each property called “clip”, which is used to match with other property data. This is also assigned to the property address. We review the address, correct any typographical errors, and generate the full street address. Furthermore, property characteristics have been cleaned as follows. The total number of bedrooms and bathrooms for all properties was limited to a maximum of 10 to eliminate unlikely outliers. The total land area in square footage (sq. ft.), total land area in acres, and total living area were also limited to any value less than 100,000 sq. ft., 25 acres, and 6000 sq. ft., respectively, to eliminate outliers, and logged as a result of right-skew. We also have the total number of stories limited to a maximum of 4, and the total number of fireplaces limited to a maximum of 24. The year built variable ranges from 1614 to 2024, with greater density after the 1900s, as expected, with some older homes in the “First State” (Delaware). The census block group has been extracted from the 10-digit census id. Additionally, variables include a categorical variable that indicates if a property has a pool, building improvement codes, and garage type codes.

Some property attribute values were missing for a particular tax year, for example, a property had 1 bedroom in 2012, was missing in 2013, and had 1 again in 2014. This indicates a data-entry error, and therefore, we apply an imputation method to fill in the missing values. We use forward and backward imputation to fill missing values, for example, by imputing a 2013 value from 2012 or 2014. After imputing several variables[1] and performing additional cleaning, the final historical property dataset comprised 4,136,259 observations.

3.2.2 Owner transfer data

The second dataset about property sales, the owner-transfer dataset, is a unique transaction-level cross-section, and repeat-sales dataset. This dataset also contains the unique identification number assigned to each property called “clip”, the property sales date (the sale-derived recording date), the sale amount, type of property (e.g., single-family residence, condominium, commercial, duplex, apartment), and an interfamily-related indicator. The initial dataset contained property transactions that dated from 1901 to 2025 for a total of 1,072,166 observations; however, 423,203 of these observations were dropped due to not having a recorded sale amount. Properties were also limited to single-family residences (386,345 observations dropped), and non-arm’s length transactions were dropped from the data (46,968 observations dropped). Property sale amounts were limited to any property sales ranging between \$40,000 and \$1,000,000 to eliminate outliers. After this cleaning, the final owner-transfer datasets consisted of properties sold between 1973 and 2025, with a total of 355,190 transactions.

[1]We imputed the following variables: total number of bedrooms, total number of bathrooms, total land area (sq. ft. and acers), parcel level latitude and longitude, property-built year, census id, total living area (sq. ft.), total number of stories, building improvement codes, and zip code.

3.2.3 Permit data

Lastly, the permit data provides details about what type of building permit was acquired for construction, renovation, and demolition for residential and commercial properties across the U.S. This dataset also includes "clip", permit effective date (1981 to 2025), permit types, and permit description. It has a total of 6,86,963 permit records, organized by cross-section and panel format, as properties obtain single or multiple permits over the period for various purposes. The property generally requires permits for bedroom or bathroom renovations, HVAC, kitchen remodels, new construction, and roofing, which are important to control in the solar hedonic model to capture the true solar effect. We perform a textual analysis of the permit description and categorize it as follows[2]: bathroom renovation, bedroom renovation, structural work, kitchen remodel, major addition, repair, roof renovation, and utility work.

3.3 Match property and solar installation data

All three property datasets have been matched using the unique identifier "clip" available in the datasets. First, we merged "owner transfer" and "historical property" data using the "clip" and year variables (sale year for owner transfer, tax year for historical data) for the years 2009 to 2024. As there was insufficient historical property data for before 2009, we used the 2010 tax year as a proxy to merge with those owner transfer years in which the property was sold before 2009. This helps us observe the market prior to 2009, but we are unable to capture renovations to properties that may have occurred during that period[3]. The final, cleaned, merged dataset consisted of 290,561 observations spanning from 1974 to 2025. As we focus our spatial analysis on the years 1998 to 2025, dropping any years outside of this range brings the count of observations down to 209,328 individual home transactions. Furthermore, we match this home transaction data with permit data using "clip", which yields 271,542 individual home transactions.

Finally, we match both solar installation and property data to investigate the relationship between solar installation and house price premium. We match the solar system installation data with property data using the individual property address. Initially, we were able to match around 60% of DE solar homes with property data; however, we also checked for typo errors in the addresses for the remaining 40% of data. This process increases the matching percentage by 4%, therefore, we have 64% of solar homes that match with property data.

We have a similar kind of data structure that was used by (Dastrup et al., 2012) to estimate the relationship between solar installation and property values. We assign property transactions into five different categories as follows: (1) property has been sold after solar

[2] Permit categories were assigned using a keyword-based text classification procedure implemented in Python using the "scikit-learn" machine learning library (Pedregosa et al. 2011). Text features were extracted using the "TfidfVectorizer" class from the "sklearn.feature_extraction.text" module. A subsample of permits was then spot-checked and reviewed by a panel of three researchers to ensure that classifications were appropriate.

[3] As a robustness check, we re-estimate all models after restricting the sample to transactions from 2009 to 2025. The estimated solar effects remain positive and broadly consistent with the baseline results, including specifications with detailed renovation controls (see Appendix Table A14).

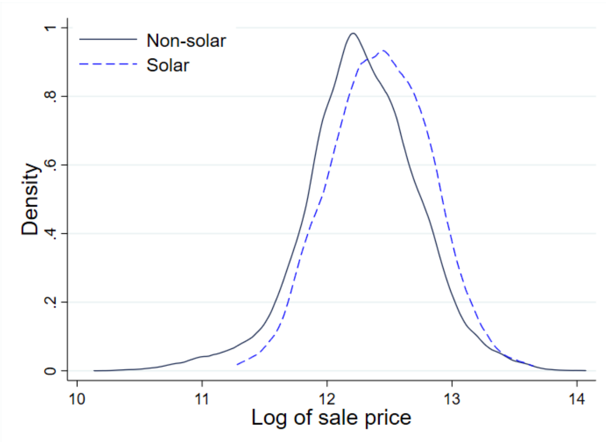
system installation (treatment group); (2) property sale and solar installed in the same year; (3) property transaction occurs prior to solar installation; (4) property has not installed solar system (control group); (5) property ever adopts solar system (i.e., categories 1,2,3 which includes all solar properties).

3.4 Descriptive statistics

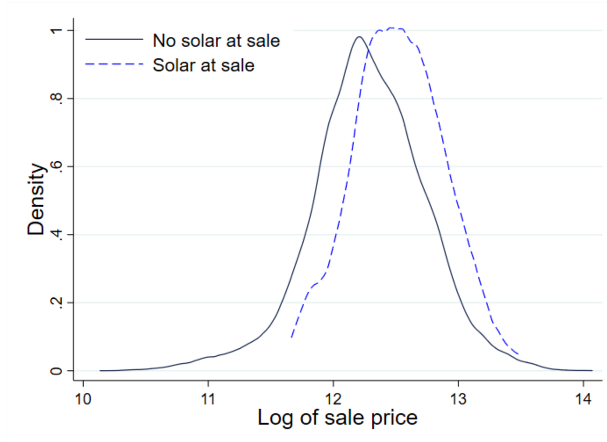
In the Appendix, Table A2 and Table A3 present descriptive statistics for residential properties in Delaware, arranged by solar adoption status. Table A2 compares homes that have ever adopted solar with those that never installed a solar system. Table A3 compares group properties sold with solar installed at the time of sale (treatment-group) and those sold without solar (control-group). We observe that solar homes sell at significantly higher prices: the average inflation-adjusted sale price for solar homes is nearly $49,000 higher than that for non-solar homes. Similarly, homes that had solar at the time of sale also have a higher price, around $50,000, than homes sold without solar. Solar homes are also larger: they have more bathrooms and bedrooms, a greater total living area, and a larger total land area; however, they are similar in the total number of stories and in the presence of a pool. On the other hand, solar homes are roughly 9-10 years newer than non-solar homes. These differences suggest that solar and non-solar homes differ systematically in observable housing characteristics. Therefore, controlling for property attributes is important in hedonic models to reduce omitted-variable bias. However, observable housing characteristics may not fully capture differences in housing quality. As discussed in sections 2.1 and 2.2, solar installation may be correlated with home renovation, which can also affect housing values. To address this concern, our analysis incorporates renovation permit records.

The primary attributes of the solar system include installed capacity (kW), system age, and ownership status. The average capacity across all in Delaware is approximately 8.1kW, while the average capacity present at the time of sale is 6.54kW. This difference likely reflects changes in system sizing over time, as more recent installations tend to have larger installed capacities than systems installed in earlier years. Solar systems are relatively young at the time of sale, with an average age of 5.71 years. This indicates that most systems are sold well within their expected operational lifespan. Finally, in our sample, around 73.65% of systems are owned rather than leased. However, 21.47% of homes with leased solar systems are observed at resale, compared with 8.12% of homes with owned systems, partly reflecting earlier installation of leased systems. These motivate us to explore meaningful heterogeneity in solar hedonic analysis.

Figure 1 shows the distribution of the log of property sale prices across solar, non-solar homes, and homes sold with solar. Panel A shows that homes that have ever adopted solar systems have sales prices that are right-skewed relative to non-solar homes, suggesting higher prices throughout much of the distribution rather than only at the upper tail. Similarly, Panel B shows an even greater difference when comparing homes sold with solar at the time of sale to homes sold without solar; this is consistent with the mean differences observed in Table A3. This indicates a positive relationship between solar adoption and housing sale prices, without controlling for home characteristics and housing market conditions across



(a) Solar and Non-Solar Homes



(b) Solar at time of Sale

Figure 1: Distribution of the log of property sale prices

space and time. The noticeable differences in observable characteristics indicate that simply comparing means may overestimate causal effects.

4 Empirical strategy

4.1 Hedonic model

To estimate the extent to which solar systems are capitalized into housing prices, we follow the conventional log-linear hedonic pricing approach. We begin with an empirical approach similar to that widely used in the housing and solar capitalization literature (e.g., Rosen (1974); Dastrup et al. (2012); Hoen et al. (2013)). The baseline hedonic specification is given by:

$$\log(P_{itb}) = \beta_0 + \beta_1 Solar_{it} + \gamma' Z_{it} + \eta' X_{it} + \theta' R_{it} + \tau_t + \phi_b + \varepsilon_{itb}. \tag{1}$$

where $\log(P_{itb})$ is the natural logarithm of the inflation-adjusted sale price for property i sold in year-quarter t and census block group b. $Solar_{it}$ is the treatment variable and equals 1 if the property had a solar system at the time of sale, and 0 otherwise. The vector Z_{it} captures observable solar system attributes, including: (i) solar system capacity in kilowatts (kW), (ii) an indicator for whether the property ever adopted a solar system, and (iii) solar system age at the time of sale. The vector X_{it} contains standard housing and lot characteristics commonly used in hedonic models, including total number of bedrooms and bathrooms, total living area in square feet (sq. ft.), property age, an indicator for whether the property has a pool, total number of stories, and lot area in acres. Property age is constructed using the inverse hyperbolic sine (IHS) transformation because some properties have zero age. These variables control for observable differences in structural housing attributes.

To address potential confounding from recent home improvements, R_{it} includes a set of renovation indicators derived from permit data. Each indicator equals one if the property underwent the corresponding improvement within five years preceding the sale, including structural work, bathroom renovation, bedroom renovation, kitchen renovation, major addition, repair, roof renovation, and utility work. Including these controls helps isolate the capitalization of solar systems from contemporaneous quality upgrades that may otherwise bias the estimates. τ_t is year-quarter fixed effects, which absorb common temporal shocks to the housing market, including macroeconomic conditions, housing cycles, broader time trends, and seasonality in transactions. ϕ_b is the census block-group fixed effects, which capture time-invariant neighborhood characteristics such as baseline location desirability, access to amenities, and persistent socioeconomic conditions.

We cluster standard errors at the census block-group level to account for spatial correlation in unobserved shocks affecting transactions within the same neighborhood over time. The coefficient of interest is β_1[4], which represents the average price premium associated with having solar installed at the time of sale. The vectors γ', η', and θ' contain the coefficients for solar attributes, property characteristics, and renovation controls, respectively. ε_{itb} is an i.i.d. error term.

4.2 Repeat-sales model

To further address concerns regarding unobserved housing quality and selection bias in solar adoption, we supplement the cross-sectional hedonic analysis with a repeat-sales model that exploits within-property variation over time. The repeat-sales specification is given by:

$$\log(P_{it}) = \beta_0 + \beta_1 Solar_{it} + \gamma' Z_{it} + \eta' X_{it} + \theta' R_{it} + \tau_t + \phi_i + \varepsilon_{it}, \tag{2}$$

In this specification, we replace the census block-group fixed effect (ϕ_b) with a parcel-level (property) fixed effect (ϕ_i), which absorbs all time-invariant property characteristics, including baseline structural quality, lot features, micro-location attributes, and persistent neighborhood characteristics. Identification is therefore restricted to properties that transact more than once during the sample period, with estimation driven by changes in sale prices within the same property over time (Cannaday et al., 2005; Hallstrom and Smith, 2005; Harding et al., 2007; Angrist and Pischke, 2009; Yu et al., 2021). By differencing out all time-invariant property attributes, the repeat-sales model substantially reduces omitted-variable bias arising from unobserved heterogeneity that may confound cross-sectional hedonic estimates. Standard errors are clustered at the parcel level to account for serial correlation in transaction prices within the same property.

A key limitation of conventional repeat-sales models is the implicit assumption that housing quality and structural attributes remain constant across transactions of the same property. In practice, housing quality may change discretely over time due to renovations, additions, remodeling, or other structural upgrades. This issue is particularly relevant in

[4] The coefficient β_1 is used to calculate the solar premium as a percentage of the property price. In a log-liner model, the percentage contribution (pc) is calculated as, pc = $\exp(\beta_1) - 1$ (Halvorsen and Palmquist, 1980; Giles, 1982).

the context of rooftop solar adoption, which may coincide with roof replacement, electrical upgrades, HVAC improvements, or major home remodeling. If such improvements are unobserved, repeat-sales estimates may incorrectly attribute resulting price changes to solar adoption rather than to other contemporaneous investments in housing quality.

We address this concern by incorporating historical property characteristics and permit-based renovation controls into both the hedonic and repeat-sales models. These controls capture the timing and type of renovation activity occurring between transactions, allowing us to account for observable quality changes that violate the constant-quality assumption underlying standard repeat-sales models. Specifically, renovation indicators include structural work, bathroom renovations, bedroom renovations, kitchen renovations, major additions, repairs, roof renovations, and utility-related improvements occurring within five years prior to the property sale.

Permit records thus provide a time-stamped proxy for quality upgrades that are otherwise unobserved or measured with error in administrative housing datasets. Incorporating these controls strengthens identification in the repeat-sales model by separating the value associated with solar adoption from the value generated by other property improvements. Under the identifying assumption that, conditional on year-quarter fixed effects and observed renovation activity, the remaining within-property shocks to housing prices are orthogonal to the timing of solar adoption, the coefficient β_1 can be interpreted as the causal capitalization effect of having a solar system installed at the time of sale.

5 Results and discussion

5.1 Hedonic and repeat-sales estimates

The preliminary estimated results for the relationship between solar adoption and the home price premium are presented in Table 1, using both hedonic and repeat-sale models. Model 1 presents results for a specification without housing controls and fixed effects (F.E.). The estimated coefficient indicates that homes with solar systems sell for 28.14% higher sale prices than non-solar homes. This large estimate likely shows some selection bias, capturing not only the effect of the solar system but also differences in housing characteristics, neighborhood quality, and market timing. However, Model 2 shows a substantial reduction in the estimated effect of sale price to 13.31% when we control for spatial and temporal controls using census block group and year-quarter fixed effects. This indicates that solar homes tend to be located in higher-valued neighborhoods and sold during good market conditions. The solar effects remain economically large, suggesting that solar adoption is positively capitalized in local housing markets.

Furthermore, we incorporate detailed housing characteristics and solar system attributes into the hedonic model with census block-group and year-quarter FEs. Model 3 shows a positive and significant effect 7.42%, which is smaller than prior models. However, Model 4 shows a slightly higher impact of solar 12.97%, with a negative effect of solar age, and a

Table 1: Estimated results using hedonic and repeat-sales models

VARIABLES	Dependent Variable: Log of Property Sale Price					
	Hedonic Model				**Repeat-Sale Model**	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar at time of sale	0.248***	0.125***	0.0715***	0.122***	0.0456***	**0.0738*** **
	(0.0183)	(0.0146)	(0.0112)	(0.0186)	(0.0119)	(0.0202)
Solar capacity			0.00212**	0.00201**	-0.0118	-0.0125
			(0.000966)	(0.000965)	(0.0138)	(0.0137)
Solar homes (ever)			0.00578	0.00673	0.151	0.159
			(0.00908)	(0.00909)	(0.151)	(0.150)
Solar age				-0.00923***		-0.00565*
				(0.00232)		(0.00333)
Total number of bedrooms			0.0208***	0.0208***	-0.00693	-0.00694
			(0.00344)	(0.00344)	(0.0173)	(0.0173)
Total number of bathrooms			0.0949***	0.0949***	0.154***	0.154***
			(0.00362)	(0.00362)	(0.0138)	(0.0138)
Ln(Total living area, sq. ft.)			0.275***	0.275***	-0.00923	-0.00926
			(0.0113)	(0.0113)	(0.00702)	(0.00702)
Property age (IHS)			-0.0602***	-0.0602***	-0.00341***	-0.00342***
			(0.00265)	(0.00265)	(0.00113)	(0.00113)
Property with pool			0.0757***	0.0756***	0.00276	0.00268
			(0.0148)	(0.0147)	(0.0232)	(0.0232)
Total number of stories			-0.0114	-0.0115	0.0358**	0.0358**
			(0.00718)	(0.00718)	(0.0155)	(0.0155)
Ln(Total land area, acres)			0.120***	0.120***	0.0227**	0.0227**
			(0.00448)	(0.00448)	(0.0108)	(0.0108)
Constant	12.26***	12.26***	10.30***	10.30***	11.92***	11.92***
	(0.00104)	(4.71e-05)	(0.0761)	(0.0761)	(0.0845)	(0.0844)
Observations	272,217	272,196	271,542	271,542	176,737	176,737
R-squared	0.001	0.561	0.695	0.695	0.854	0.854
Census Block FE	×	✓	✓	✓	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the log of home sale price, and the treatment variable is the presence of solar at the time of sale. Standard errors are clustered at the census block group and parcel levels in models 2-4 and 5-6, respectively. In model 6, estimates suggest that having solar at the time of sale increases property sale price by 7.65% ($22,379.24), after controlling for solar attributes and basic property characteristics.

positive effect of solar capacity. These estimates represent solar home sale price premiums of \$21,656.57 and \$37,912.82 relative to non-solar homes, respectively (See detailed capitalization calculation in section A.6). The consistently positive and significant coefficients across models confirm that solar systems affect housing values positively in Delaware.

Incorporating solar system attributes helps to understand the mechanisms underlying capitalization. This motivates a detailed investigation of heterogeneity in solar attributes. Solar system capacity has a positive effect on sale prices, indicating that prospective home buyers value larger systems for greater electricity savings. In contrast, solar age has a negative effect on the sale price, indicating that older systems reduce the sale price. We investigate this further in sections 5.3 and 5.5.

Columns 5 and 6 present estimated results using the repeat-sales model approach. In this approach, solar has been installed between two transactions, before and after the installation of solar systems. This helps to explore within-property variation over time using parcel-level fixed effects and controls for all time-invariant property characteristics. The estimated results show that the solar home price increases by 4.66% to 7.65% and that there is a negative solar age effect, consistent with the hedonic model. These estimates indicate increases of \$13,632.31 to \$22,379.24 in solar home sale prices. Although smaller than the hedonic estimates, these effects remain statistically significant, economically meaningful, and consistent with prior literature (Dastrup et al., 2012; Hoen et al., 2013; Best et al., 2021).

This decrease in magnitude compared to the hedonic model is expected. By comparing the same property before and after solar adoption, the repeat-sales model mitigates bias arising from unobserved, time-invariant differences in housing quality. As a result, these estimates provide stronger evidence of a causal effect of solar adoption, rather than selection into higher-quality homes, as suggested by the descriptive statistics. However, prior literature (Dastrup et al., 2012; Gillingham and Watten, 2024) suggests that these estimates may still be subject to omitted variable bias, including time-varying home improvements and demographic characteristics that influence sale prices. In the next section, we address this concern by incorporating detailed home improvement information using renovation permit data, allowing us to address this omitted variable bias problem and estimate the true capitalization effects of solar adoption.

As an additional robustness check, we conduct a series of placebo tests that randomly assign solar treatment to non-solar homes while preserving key features of the observed solar-treatment distribution. Across these placebo specifications, the estimated solar effects are statistically insignificant, providing additional evidence that the premiums estimated in our main specifications are not driven by spurious correlations (see Appendix Table A17).

5.2 Hedonic and repeat-sales results with renovation controls

In this section, we present the hedonic and repeat-sales models, including renovation attributes to account for major changes made to the home within five years prior to the property sale. Including renovation controls allows us to isolate the capitalization of solar systems from major housing quality upgrades that are likely correlated with solar adoption.

We start with a baseline specification without fixed effects in Table 2 , in Model 1. The estimated coefficient indicates that the solar home price increases by 22.87% with renovation indicators (e.g., Structural work, Bathroom and Bedroom renovation, Kitchen renovation, Major addition, Repair, Roof renovation, and Utility work), which is 5.27 percentage points lower than the corresponding baseline estimate in Table 2, (relative reduction by 18.7%), Model 1. This suggests that not accounting for recent home improvements results in an upward bias in solar capitalization.

In Models 2 and 3, we include census block group and year-quarter fixed effects, along with solar attributes, basic property characteristics, and renovation controls. Under this specification, the estimated solar effect remains positive and statistically significant at 9.14% after controlling for basic property characteristics and renovation in Model 2. Similarly, in model 3, the estimated solar effect is much higher, 12.07%, and solar capacity and solar age are consistent with the prior estimates.

However, the repeat-sales specifications in models 4 and 5 provide the cleanest estimates of solar capitalization. After accounting for renovations and other time-varying characteristics, having solar at the time of sale increases property sale price by 4.06% to 5.68%, after controlling for solar attributes, basic property characteristics, and renovation controls. This implies that the true solar price premium ranges from $11,863.67 to $16,613.22 at the average solar home sale price. These estimates align closely with prior findings from high-penetration solar markets such as California, Massachusetts, Hawaii, and Arizona, suggesting that solar capitalization is robust across diverse markets, even in low-penetration solar markets, when renovation activity is controlled properly (Dastrup et al., 2012; Hoen et al., 2013; Wee, 2016; Hoen et al., 2017; Qiu et al., 2017; Gillingham and Watten, 2024).

We also find significant correlations between home sale price and renovation controls across all models, except for bathroom renovation in the hedonic model. The renovation estimates are economically large and statistically significant at the 1% level, underscoring the importance of controlling for changes in housing quality. Structural work, Bathroom and Bedroom renovation, Kitchen renovation, Major addition, Repair, Roof renovation, and Utility work are all associated with an increase in sale prices, particularly in repeat-sales models. For example, as expected, Models 4 and 5 show that major additions, roof renovations, and bedroom renovations increase home sale prices by 6.02%, 9.38%, and 9.19%, respectively. These findings strongly suggest that solar adoption coincides with broader reinvestment in housing quality, rather than occurring in isolation.

As discussed in prior literature, home updates occurring around the same time as solar adoption may affect home prices (Dastrup et al., 2012; Gillingham and Watten, 2024). For example, a household may install a heat pump or do other home improvements between two transactions, or they may replace the roof before installing a solar system. One important point to note is that solar homes have not only experienced higher home sale prices but have also benefited from additional improvements that could further increase their selling prices.

Table 2: Estimated results using hedonic and repeat-sales models with renovation controls

VARIABLES	**Dependent Variable: Log of Property Sale Price**				
	Hedonic Model			**Repeat-Sale Model**	
	Model 1	Model 2	Model 3	Model 4	Model 5
Solar at time of sale	0.206***	0.0875***	0.114***	0.0398***	**0.0553*** **
	(0.0179)	(0.0105)	(0.0179)	(0.0123)	(0.0193)
Solar capacity			0.00174*		-0.0139
			(0.000962)		(0.0136)
Solar homes (ever)			0.00817		0.177
			(0.00908)		(0.148)
Solar age			-0.00814***		-0.00389
			(0.00230)		(0.00331)
Total number of bedrooms		0.0205***	0.0204***	-0.0150	-0.0151
		(0.00342)	(0.00342)	(0.0167)	(0.0167)
Total number of bathrooms		0.0952***	0.0953***	0.128***	0.128***
		(0.00361)	(0.00361)	(0.0134)	(0.0134)
Ln(total living area, sq. ft.)		0.275***	0.275***	-0.00484	-0.00476
		(0.0113)	(0.0113)	(0.00698)	(0.00698)
Property age (IHS)		-0.0568***	-0.0567***	0.00331***	0.00331***
		(0.00275)	(0.00275)	(0.00121)	(0.00121)
Property with pool		0.0737***	0.0738***	0.00121	0.00122
		(0.0147)	(0.0148)	(0.0234)	(0.0234)
Total number of stories		-0.0113	-0.0112	0.0304**	0.0311**
		(0.00719)	(0.00720)	(0.0154)	(0.0155)
Ln(total land area, acres)		0.119***	0.119***	0.0193*	0.0201*
		(0.00447)	(0.00447)	(0.0109)	(0.0109)
Structural work	0.250***	0.0250***	0.0251***	0.0644***	0.0646***
	(0.00773)	(0.00696)	(0.00698)	(0.00742)	(0.00742)
Bathroom renovation	-0.0300	0.0173	0.0174	0.119***	0.119***
	(0.0450)	(0.0277)	(0.0276)	(0.0417)	(0.0417)
Bedroom renovation	0.0639***	0.0372***	0.0370***	0.0885***	0.0883***
	(0.0205)	(0.0120)	(0.0121)	(0.0211)	(0.0211)
Kitchen renovation	-0.0351**	0.0464***	0.0464***	0.222***	0.222***
	(0.0163)	(0.00999)	(0.00998)	(0.0180)	(0.0180)
Major addition	0.309***	0.0515***	0.0514***	0.0585***	0.0586***
	(0.00408)	(0.00564)	(0.00565)	(0.00401)	(0.00401)
Repair	0.0827**	0.0333*	0.0330*	0.0641**	0.0640**
	(0.0333)	(0.0173)	(0.0173)	(0.0258)	(0.0258)
Roof renovation	0.0834***	0.0316***	0.0296***	0.0897***	0.0896***
	(0.0144)	(0.0102)	(0.0102)	(0.0134)	(0.0134)
Utility work	0.245***	0.0237***	0.0235***	0.0580***	0.0580***
	(0.00300)	(0.00500)	(0.00500)	(0.00282)	(0.00282)
Constant	12.20***	10.28***	10.28***	11.94***	11.94***
	(0.00111)	(0.0757)	(0.0757)	(0.0829)	(0.0829)
Observations	272,217	271,584	271,542	176,763	176,737
R-squared	0.057	0.696	0.696	0.856	0.856
Census Block FE	×	✓	✓	×	×
Year-Quarter FE	×	✓	✓	✓	✓
Property FE	×	×	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the log of home sale price, and the treatment variable is the presence of solar at the time of sale. Standard errors are clustered at the census block and parcel levels in models 2-3 and 4-5, respectively. In model 5, estimates suggest that having solar at the time of sale increases property sale price by 5.68% ($16,613.22), after controlling for solar attributes, basic property characteristics, and renovation controls.

Therefore, it is important to control for home renovation.

Accounting for renovation activity, therefore, strengthens identification in two important ways. First, it prevents overestimating the solar premium by attributing home improvements to solar adoption. Second, it clarifies that the solar capitalization reflects buyers' valuation of the solar itself, by expected energy savings, and environmental benefits, rather than just overall bundled upgrades. This distinction has been largely absent from earlier solar hedonic literature due to data limitations[5].

Overall, our estimated results for repeat sales may be unbiased after controlling for renovations. By integrating home renovation controls, this study advances the literature toward more credible estimates of solar capitalization and provides valuable guidance to households in solar system investment. Furthermore, we explore the solar age effect in the next section.

5.3 Hedonic and repeat-sales models: Solar age effect

In the previous section, we observed that the solar age has a negative impact on home values. In this section, we examine in more detail solar capitalization by solar system age, which has received limited empirical attention in the prior literature. To explore this heterogeneity, we replace the continuous solar age variable with mutually exclusive solar age indicators[6], computed as the difference between the solar installation date and the property sale date.

Table 3, Model 1 to 3, the hedonic model shows a monotonic decline in capitalization as the age of solar systems increases. Homes with new solar systems installed within three years prior to sale exhibit the largest price premium, 9.80% to 12.86%, even after controlling for housing characteristics and renovation. Solar systems installed four to six years prior to sale remain positive and statistically significant, though the magnitude is reduced to 6.72% to 8.92%. Similarly, for solar systems installed seven to nine years prior to sale, effects remain consistent, with a range of 6.08% to 7.79%. For older systems, particularly those ten years or more since installation, estimated premiums decline substantially and are statistically insignificant.

[5](Dastrup et al., 2012) use building permit data, and (Gillingham and Watten, 2024) use Multiple Listing Service (MLS) information from CoreLogic to control for roof replacement, which may coincide with solar installation. This relationship is plausible because rooftop solar systems are long-lived investments, and homeowners with aging roofs may replace the roof before installing solar to avoid the future cost of removing and reinstalling the panels when roof replacement becomes necessary. This consideration is particularly relevant because asphalt/composition roofs and rooftop solar systems have relatively long and potentially overlapping service lives, approximately 25-30 years (Dunnigan 2020; Walker and McDevitt 2025). In our dataset, 68 of 4,344 solar homes (1.57%) had a roof replacement around the time of or prior to solar installation, compared with 1,360 of 161,357 non-solar homes (0.84%) with a roof replacement prior to sale. Although roof replacement is uncommon in both groups, the higher rate among solar homes is consistent with some homeowners coordinating roof replacement with solar installation.

[6]The solar age indicator is defined as follows: solar age (0-3) = 1 if solar age is 0-3 years old, and 0 otherwise; solar age (4-6) = 1 if solar age is 4-6 years old, and 0 otherwise; solar age (7-9) = 1 if solar age is 7-9 years old, and 0 otherwise; and solar age ($\geq$10) = 1 if solar age is 10 years old or more, and 0 otherwise.

However, the Model 4 to 6 repeat-sales model provides a cleaner estimate of the solar system age. These results indicate that capitalization is strongest for newer systems and declines sharply with age. Solar systems installed within three years of home sale increase home value by 4.62% to 6.15%, whereas systems aged four to six years modestly increase home value by 3.94% to 4.42%. In contrast, systems older than six years do not produce statistically significant effects on sale prices. The consistency of the solar age effect across both specifications supports the interpretation that solar capital depreciates, rather than unobserved housing quality, drives the observed pattern.

Table 3: Estimated results using hedonic and repeat-sales models: Solar age effect

	Dependent Variable: Log of Property Sale Price					
VARIABLES	**Hedonic Model**			**Repeat-Sales Model**		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar age 0–3 years (N=335)	0.121***	0.100***	0.0935***	0.0597***	0.0574***	0.0452***
	(0.0154)	(0.0154)	(0.0152)	(0.0155)	(0.0155)	(0.0152)
Solar age 4–6 years (N=278)	0.0855***	0.0658***	0.0651***	0.0433**	0.0387**	0.0405**
	(0.0150)	(0.0153)	(0.0152)	(0.0220)	(0.0194)	(0.0198)
Solar age 7–9 years (N=163)	0.0751***	0.0557***	0.0591***	0.0191	0.0149	0.0261
	(0.0216)	(0.0213)	(0.0213)	(0.0231)	(0.0233)	(0.0238)
Solar age $\geq$ 10 years (N=100)	0.0271	0.00816	0.0113	0.000605	-0.00698	-0.0138
	(0.0199)	(0.0203)	(0.0202)	(0.0389)	(0.0399)	(0.0407)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,584	271,542	271,542	176,763	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block Group FE	✓	✓	✓	×	×	×
Year-Quarter FE	✓	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓
Property Controls	✓	✓	✓	✓	✓	✓
Solar Attributes	×	✓	✓	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Solar age groups indicate the age of the solar system at the time of sale. Standard errors are clustered at the census block-group level in Models (1)–(3) and at the parcel level in Models (4)–(6). The complete table presented in Appendix Table A11.

The prospective solar home buyers capitalize on the expected net present value of remaining electricity bill savings. Newer systems offer greater expected future generation, no maintenance risk, and longer warranty coverage, which increases buyer willingness to pay for a solar home. Conversely, older systems face a decrease in panel efficiency, inverter replacement costs, and uncertainty in long-term performance, leading buyers to discount the home value. Importantly, older systems seven or more years are not purely symbolic, as they are not fully capitalized. However, the near-full capitalization within the first three years of solar installation suggests that homeowners who sell relatively soon after adopting solar are more likely to recover their upfront investment through a home resale premium.

These findings contribute to the emerging literature on solar hedonics by demonstrating that capitalization is highly time-dependent. The solar age effect has not been explored in prior literature, which typically reports average capitalization effects across regional housing markets (Dastrup et al., 2012; Hoen et al., 2013; Ma et al., 2016; Wee, 2016; Hoen et al., 2017; Qiu et al., 2017; Lan et al., 2020; Begley and Hoen, 2021; Best et al., 2021, 2023; Gillingham and Watten, 2024; Asproudis et al., 2024).

5.4 Policy counterfactual simulation of rooftop solar premium

The above empirical analysis estimates the extent to which rooftop solar systems are capitalized into residential property values. However, property resale premiums represent only one component of the private return to solar investment. Homeowners also benefit from electricity bill savings, solar credits (compensation) for excess electricity exported to the grid via net-metering programs, revenues from Solar Renewable Energy Credits (SRECs), and reductions in upfront installation costs through federal ITC and state incentive Green Energy Grant (GEG) programs. Table 4 summarizes the principal federal and state policies affecting these returns through different channels and at different times in the system life span.

Table 4: Federal and Delaware State Policies Supporting Residential Rooftop Solar

Policy	**Established Period (Current Status)**	**Incentive or Compensation**	**Timing of Benefit**
Federal			
Residential Clean Energy Credit (Federal ITC)	Effective 2005–2025; terminated after Dec. 31, 2025	30% of qualified solar installation expenditures; no overall dollar cap	Claimed through federal income tax return following installation
Delaware			
Green Energy Grant (GEG)	Green Energy Program established in 1999; incentive levels vary over time; ongoing	Capacity-based installation grant; historical incentive used in the analysis: \$0.80/W	At or following installation
Solar Renewable Energy Certificates (SRECs)	Renewable Portfolio Standard framework established in 2005; ongoing	1 SREC per 1,000 kWh of qualifying solar generation; value determined by applicable procurement/market arrangements	Earned over time as electricity is generated
Net metering / Export Compensation	Net-metering framework established in 1999; subject to statutory and utility rules; ongoing	Bill credits for eligible excess solar generation exported to the grid	Realized through electricity bills over time
Low- to Moderate-Income (LMI) Solar Program	LMI Solar Pilot program launched in 2022; ongoing	Low-income: cost-free system up to 4 kW; moderate-income: program covers 70% of cost and homeowner pays 30%, up to 6 kW	At installation

Note: The table summarizes major federal and Delaware policies affecting the private returns to residential rooftop solar. Policy parameters and incentive levels have changed over time. The counterfactual simulations incorporate the policies applicable to the representative homeowner and vary selected policy instruments individually. See Appendix subsection A.4 for detailed policy histories, eligibility requirements, and parameter values used in the simulations.

To evaluate these jointly, we develop a series of policy counterfactual simulations that combine the estimated property resale premium with electricity bill savings and policy-dependent financial benefits over the system's expected 30-year lifespan. The simulations address three questions. First, how does the composition of homeowner value evolve as a solar system ages? Second, how do alternative federal, state, and utility policy instruments affect homeowners' returns? Finally, how long must homeowners retain the system before cumulative benefits exceed the initial investment under different policy environments? This procedure allows us to compare how policies that reduce upfront costs and policies that provide benefits over the system's lifetime affect homeowners' investment returns.

5.4.1 Simulation framework

The simulation combines the estimated solar resale premium with the major financial benefits homeowners receive from rooftop solar. A homeowner may realize value through three channels: electricity-bill savings, revenues from SRECs, and the solar premium capitalized into the property price when the home is sold. The initial investment cost is adjusted according to the federal ITC and Delaware GEG available under each policy scenario. This framework therefore evaluates the overall private return to rooftop solar rather than focusing on the resale premium alone.

For a homeowner who installs a solar system in year 0 and sells the property in year T, where $T = 1, \dots, 30$, total homeowner value is defined as

$$V_T = PV(BS_T) + PV(SREC_T) + PV(RP_T), \tag{3}$$

where $PV(BS_T)$ is the present value of cumulative electricity-bill savings realized from solar through year T, $PV(SREC_T)$ is the present value of SREC revenues received over the same period, and $PV(RP_T)$ is the estimated remaining solar resale premium realized when the property is sold in year T. More explicitly,

$$V_T = \sum_{t=1}^{T} \frac{BS_t}{(1+r)^t} + \sum_{t=1}^{T} \frac{SREC_t}{(1+r)^t} + \frac{RP_T}{(1+r)^T}, \tag{4}$$

where r denotes the discount rate. Electricity-bill savings in year t, BS_t, depend on annual solar generation by season, household electricity use, retail electricity prices, and the compensation received for excess electricity exported to the grid:

$$BS_t = SC_t P_t^{\text{retail}} + EX_t P_t^{\text{export}}, \tag{5}$$

where SC_t is solar electricity self-consumed by the household, EX_t is excess solar electricity exported to the grid, P_t^{retail} is the retail electricity price, and P_t^{export} is the applicable export compensation rate. Annual solar generation is allowed to decline over time according to the assumed panel degradation rate.

Similarly, SREC revenues depend on annual electricity generation and the applicable value of SRECs:

$$SREC_t = \frac{G_t}{1{,}000} P_t^{\text{SREC}}(1-a), \tag{6}$$

where annual solar generation in year t is given by

$$G_t = G_1(1-d)^{t-1}. \tag{7}$$

Here, G_1 denotes first-year solar generation in kWh, d is the annual panel degradation rate, P_t^{SREC} is the SREC price per credit, and a denotes any aggregator or transaction fee. Because one SREC is earned for each 1,000 kWh of qualifying solar generation, annual SREC revenue depends directly on system output.

The resale premium, RP_T, is derived from the solar-age estimates from the preferred repeat-sales model. It represents the remaining value of the solar system capitalized into the property's transaction price if the homeowner sells in year T.

We define net private benefits at the time of sale as total homeowner value less the homeowner's net initial investment:

$$NB_T = V_T - C_0^{\mathrm{net}}, \tag{8}$$

where C_0^{net} denotes the homeowner's out-of-pocket installation cost after applicable upfront incentives. Under a policy scenario with both the federal ITC and Delaware GEG,

$$C_0^{\mathrm{net}} = C_0 - ITC - GEG, \tag{9}$$

where C_0 is the gross installation cost. Under counterfactual scenarios, the relevant policy component is removed or modified. For example, eliminating the federal ITC increases C_0^{net}, while reducing export compensation lowers future electricity-related benefits. The investment recovery year is defined as the first year in which cumulative net benefits become nonnegative: $T^* = \min\{T : NB_T \geq 0\}$.

We evaluate homeowner outcomes for each possible year of property sale over an assumed 30-year system lifespan. Counterfactual scenarios change one policy component at a time while holding the physical characteristics of the representative solar system and other assumptions constant. This approach allows us to identify how federal tax incentives, state grants, SREC revenues, and electricity export compensation affect both the magnitude of homeowner returns and the timing of investment recovery.

5.4.2 Evolution of homeowner value over the solar system life cycle

We first focus on the two core sources of homeowner value: the present value of cumulative electricity bill savings realized during ownership and the solar premium realized at property resale, as shown in Figure 2. This allows us to illustrate how the composition of solar value changes over the system's life before introducing policy-dependent benefits, including SRECs, state grants, and alternative export compensation, in the counterfactual simulations in Figure 3.

In Figure 2, the dotted brown line represents the estimated present value of the solar home resale premium, while the dashed teal line represents the cumulative electricity-bill savings under the medium scenario. The solid blue line combines these two components to measure total homeowner value at each possible year of sale. The green line reports net benefits after subtracting the net installation cost of the system, and the shaded regions represent the range between the low- and high-scenario assumptions.

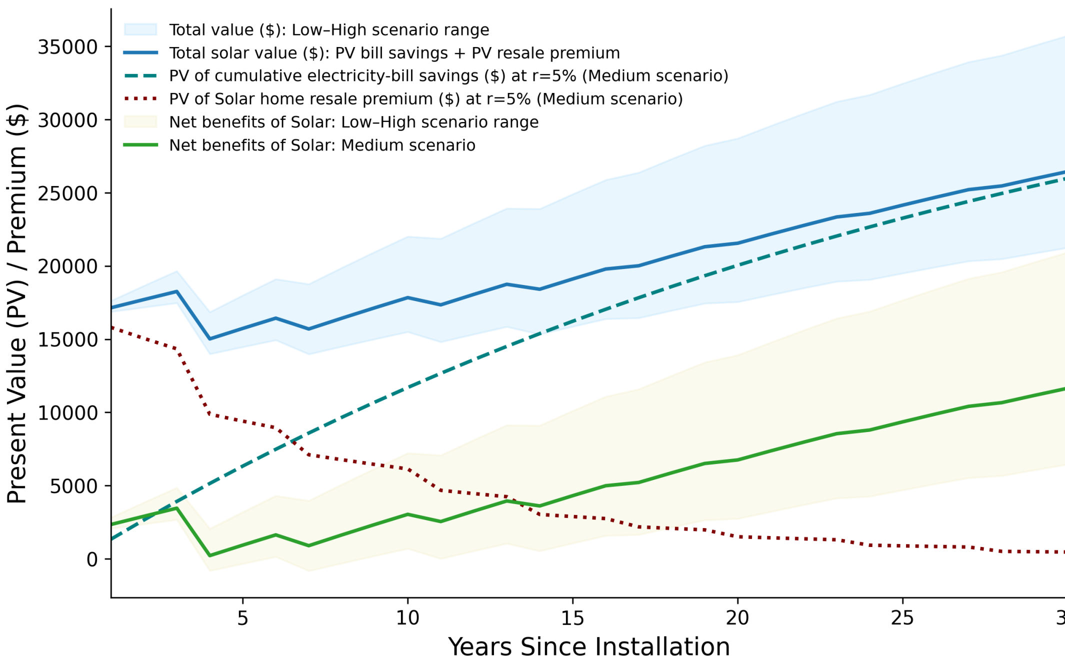


Figure 2: Present value of residential rooftop solar benefits over the system lifetime

Notes: [a]The figure illustrates the evolution of the economic value of an average 6.5-kW residential rooftop solar system over a 30-year lifespan. Three scenarios (Low, Medium, and High) are considered based on alternative assumptions regarding electricity generation and electricity prices. (1) Annual electricity generation G_t is assumed to decline at a panel degradation rate of $d = 0.5\%$. Initial generation is assumed to be about 8,750 kWh in the first year in the medium scenario. It is calculated as: $G_t = G_1(1-d)^{t-1}$ in year t. (2) Residential electricity prices are assumed to increase at an annual rate of $i = 2\%$, such that $P_t = P_1(1+i)^{t-1}$, where $P_1 = \$0.16$/kWh. (3) Annual electricity bill savings are calculated as $S_t = G_t \times P_t$, and cumulative electricity bill savings are $CS_t = \sum_{j=1}^{t} S_j$. (4) The present value of annual savings is calculated as $PV(S_t) = S_t/(1+r)^t$, while cumulative discounted savings are $PV(CS_t) = \sum_{j=1}^{t} S_j/(1+r)^j$ (dashed teal line), using a baseline discount rate of $r = 5\%$. (5) Solar home resale premiums (dotted brown line) are derived from the estimated solar-age capitalization effects obtained from the repeat-sales model and are discounted using the same discount rate ($r = 5\%$). (6) Total economic value (blue line) is defined as the sum of discounted cumulative electricity-bill savings and discounted resale premiums. (7) Net benefits (green line) are calculated by subtracting the average net installation cost of a 6.5-kW solar system in Delaware ($14,781 after incentives). The solid lines represent the medium scenario, while the shaded regions indicate the range between the low and high scenarios based on alternative assumptions regarding electricity generation and electricity prices.

[b]Residential electricity prices are assumed to increase at an annual rate of 2%, consistent with long-run projections from the U.S. Energy Information Administration (EIA, 2025b). The baseline electricity price is assumed to be $0.16/kWh, reflecting the average residential electricity price reported for Delaware by the EIA in 2024 (EIA, 2025a). Low and high scenarios assume electricity prices of $0.14/kWh and $0.20/kWh, respectively.

The results show that resale capitalization is strongest early in the system's life. During the first three years following installation, the estimated resale premium is approximately $14,000 to $16,000, implying that homeowners recover roughly 100–115% of the average net installation cost if the property is sold within this period (see Section A.6). Thus, homeowners who move relatively soon after installation may recover a substantial porting, or potentially all of their initial investment through the housing market rather than relying exclusively on future electricity savings. The resale premium subsequently declines with system age. By approximately years 4 to 6, the premium falls to about \$9,000-\$12,000, corresponding to roughly 80% of the original system cost. This pattern is consistent with buyers placing less value on older systems with shorter remaining operating lives and fewer future electricity savings available to the new owner.

Importantly, the decline in resale premium does not imply a comparable decline in the overall return to the solar investment. As the homeowners remains in the property, cumulative electricity-bill savings increase and gradually replace resale capitalization as the primary source of value. Consequently, total homeowner value remains substantial even as the estimated resale premium declines, and net benefits generally increase over longer ownership periods. Therefore, Figure 2, reveals a shift in the composition of solar value over the system life cycle: resale capitalization is relatively important early in the investment, whereas realized electricity savings become increasingly important as the system ages. In other words, solar value gradually shifts from capitalized resale value to realized electricity-bill savings. This pattern helps reconcile the declining age-specific capitalization estimates with positive long-run returns to solar investment. It is also consistent with a housing market in which the resale value of solar reflects, at least in part, the remaining economic benefits available to prospective buyers.

5.4.3 Net private benefits under alternative policy scenarios

Figure 3 examines how federal, state, and utility policies affect the private financial return to rooftop solar. Net private benefits are calculated as mentioned in Equation 8. Figure 3 presents four policy environments: the current Delaware policy environment in 2026, a counterfactual with no policy support, state support without net metering, and the pre-2026 policy environment combining the federal ITC with state incentives and net metering. System characteristics and other simulation assumptions are held constant across scenarios, allowing differences in net benefits to be attributed to changes in the policy environment. The Appendix Figure A2 and Figure A3 report additional counterfactuals that separately vary grant levels, SREC treatment, and export compensation.

Figure 3 shows that policy design substantially affects both the timing of investment recovery and the magnitude of homeowner returns. The pre-2026 policy environment generates the highest benefits throughout the simulated system life. The federal ITC operates primarily through the initial investment cost: by reducing upfront costs, it substantially improves returns in the early years of system ownership. Under the current 2026 Delaware policy environment, after the federal ITC expires, net benefits are lower in the early years and become positive only around year 10. This difference matters most for homeowners who expect to sell within a relatively short period after installation, because they have less time to accumulate electricity-bill savings and other benefits before resale.

State and utility policies also contribute meaningfully to returns. Under the state-support-only scenario, net benefits remain below those under the current Delaware policy environment. The difference between these scenarios widens with system age because compensation for exported electricity contributes to the stream of operating benefits accumulated over time. In contrast, eliminating all major policy support produces the lowest returns. Under this counterfactual, net benefits remain negative for approximately the first two decades and are substantially lower even at the end of the 30-year system life. Thus, upfront incentives (ITC and GEG) primarily improve early investment recovery by reducing installation costs, whereas SREC and net-metering policies boost returns over the system's lifespan.

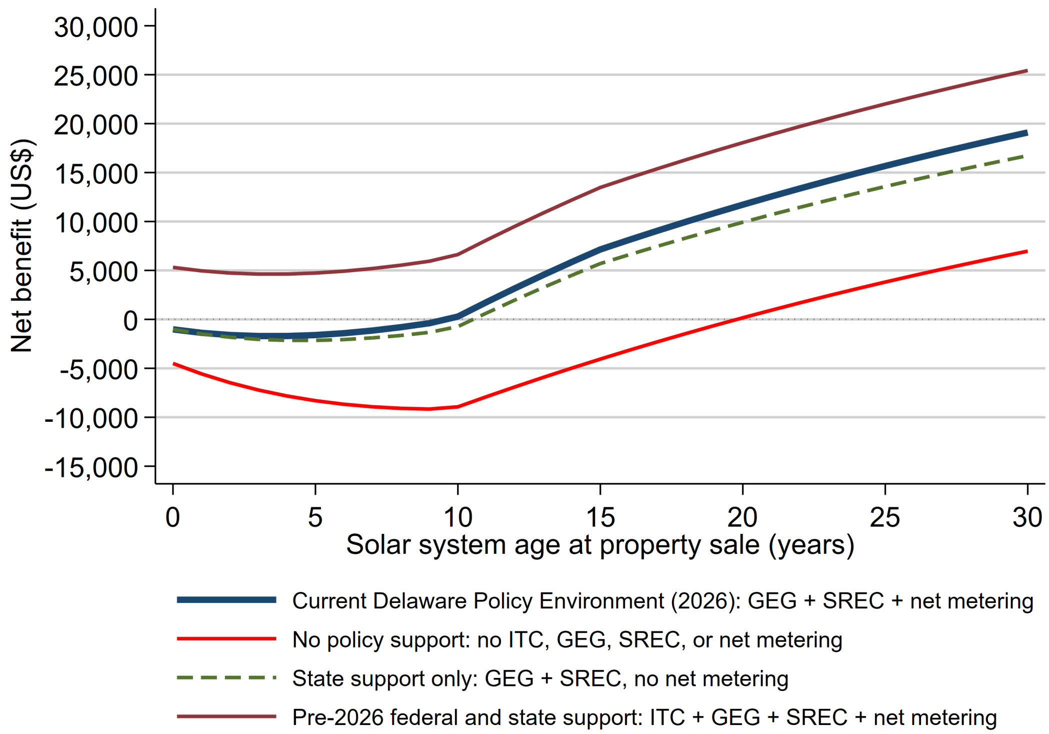


Figure 3: Net benefits under alternative solar policies over the system lifespan

Notes: The figure reports simulated net private benefits to a representative Delaware homeowner if the property is sold at different ages of the rooftop solar system over a 30-year operating life. Net benefit is defined as mentioned in Equation 8. We assume an average 6.5-kW system, cost of \$21,115.56 before federal ITC incentives, annual first-year electricity generation of 8,750 kWh, and 0.5% annual panel degradation. Household electricity consumption is 10,932 kWh annually. Electricity generation and household load are matched using monthly profiles; each month, generation first offsets contemporaneous household load, and excess generation is treated as electricity exported to the grid. The initial residential retail electricity price is \$0.1657/kWh and increases by 2% annually. Future electricity savings, SREC revenues, and the resale premium are discounted at a 5% annual real discount rate. The resale premium is based on the repeat-sales estimates reported in Table 2, model 6, and declines with system age.

These results are particularly relevant to residential mobility. A solar investment may ultimately generate positive private returns even with relatively limited policy support, but a homeowner who sells before the investment reaches break-even may face a substantially different financial outcome than one who stays in the property longer. The timing of benefits may be as important as their lifetime magnitude for household adoption decisions. The simulations therefore suggest that solar policy should be evaluated not only by its effect on lifetime returns, but also by how quickly homeowners can recover their initial investment. This consideration may be especially important in lower-penetration markets such as Delaware, where uncertainty about the financial returns to solar may remain an important barrier to adoption.

5.5 Heterogeneity by solar ownership: Owned vs leased systems

In this section, we examine solar capitalization by ownership status, distinguishing between owned systems and TPO systems, such as leased or power purchase agreement (PPA) based systems. This is important because ownership determines the allocation of residual system value, maintenance responsibilities, and lease contractual obligations during resales.

We observe heterogeneity in solar ownership and lease contract capitalization on home price premiums. Table 5, Models 1 to 3, present hedonic estimates that interact solar treatment with the solar ownership indicator. The interaction term is positive but statistically insignificant, suggesting limited power to separately identify ownership effects. The hedonic models rely on cross-property comparisons and may conflate ownership structure with unobserved, time-invariant differences across properties. However, this also shows positive and statistically significant results for the main solar effect on home values, with sales price increases of 8.70% to 11.85%.

Table 5: Estimated results using hedonic and repeat-sales models by ownership status

VARIABLES	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			Repeat-Sales Model		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar at time of sale	0.0835***	0.112***	0.106***	0.0219	0.0493**	0.0428*
	(0.0122)	(0.0160)	(0.0161)	(0.0147)	(0.0217)	(0.0220)
Solar at time of sale × Own	0.0137	0.0236	0.0198	0.0426*	0.0414*	0.0275
	(0.0199)	(0.0205)	(0.0205)	(0.0244)	(0.0230)	(0.0230)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,584	271,542	271,542	176,763	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	✓	✓	✓	×	×	×
Year-Quarter FE	✓	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓
Property Controls	✓	✓	✓	✓	✓	✓
Solar Attributes	×	✓	✓	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The treatment variable equals one if a solar system was present at the time of sale. The interaction term $Solar_{it} \times Own_{it}$ identifies whether the solar premium differs for owned systems relative to third-party-owned systems. Standard errors are clustered at the census block-group level in the hedonic models and at the parcel level in the repeat-sales models. The hedonic model is: $\log(P_{itb}) = \beta_0 + \beta_1 Solar_{it} + \beta_2 Solar_{it} \times Own_{it} + \gamma' Z_{it} + \eta' X_{it} + \theta' R_{it} + \tau_t + \phi_b + \varepsilon_{itb}$. The repeat-sales model is: $\log(P_{it}) = \beta_0 + \beta_1 Solar_{it} + \beta_2 Solar_{it} \times Own_{it} + \gamma' Z_{it} + \eta' X_{it} + \theta' R_{it} + \tau_t + \phi_i + \varepsilon_{it}$. The complete table presented in Appendix Table A10.

The repeat-sales estimates in Models 4 and 5 show positive and statistically significant effects of solar ownership at the 10% level. The estimates imply an additional premium of approximately 4 percentage points for owned compared to leased systems. The estimated resale premiums are approximately 9.49% for owned systems and 5.05% for leased systems, corresponding to around $27,740.64 and $14,765.95, respectively. However, the ownership coefficient becomes statistically insignificant in Model 6 after controlling for recent home renovations. Thus, the evidence that ownership status independently affects solar capitalization is suggestive rather than conclusive. Our results are consistent with prior literature, which suggests that leased solar systems exhibit substantially lower capitalization than owned solar systems (Begley and Hoen, 2021; Gillingham and Watten, 2024), suggesting that the effect of ownership status on capitalization is less clear in our preferred specification.

One possible explanation for the observed difference is that an owned system transfers with property without the remaining payment obligations associated with a solar lease, allowing the buyer to receive the remaining electricity-bill savings directly. In contrast, solar leased system consumers typically need to sign a 20-year term solar lease contract, meet contract-transfer requirements, and may face escalation clauses, maintenance provisions, or penalties for early termination. Although it is possible to transfer the lease contract to a new homebuyer, the buyer will assume responsibility for paying the remaining lease payments until the end of the contract, which could reduce buyers' willingness to pay. One common concern among buyers is the potential for maintenance costs or equipment obsolescence over time, or future electricity price uncertainty. These provide a potential mechanism for greater capitalization of owned systems, although the estimates do not provide strong evidence that ownership status independently explains differences in resale premiums.

5.6 Hedonic and repeat sale model: Solar capacity effect

In this section, we examine how solar capitalization varies with system size. Theoretically, a large solar system delivers higher expected electricity bill savings and may highly increase home values. But this could happen only up to the point where additional capacity exceeds the household's typical load or where buyers discount large solar systems because of aesthetics or maintenance risk. To test these intensive-margin effects, we use above-baseline homes with the interaction of treatment indicator "solar at time of sale" and solar capacity (in kW), and its quadratic term to allow for diminishing returns.

Table 6, Models 2-3, presents hedonic model estimates. The coefficient on solar capacity is small and only marginally significant, while the interaction terms (Solar at time of sale $\times$ capacity) and (Solar at time of sale $\times$ capacity)^2 are not statistically significant. However, the corresponding repeat-sales Models 4-6 show non-robust evidence that, conditional on having a solar system, larger systems sell at higher prices. These results indicate that the Delaware solar premium is an extensive-margin effect (solar home value increases more), while the intensive margin (i.e., additional kW of capacity) plays at most a secondary role.

We also run a set of robustness checks reported in Appendix Table A4 and Table A5. First, Appendix Table A4 re-estimates the models using only with (Solar at time of sale

× capacity) and (Solar at time of sale × capacity)^2, without the standard solar treatment indicator. This is treating a continuous treatment variable instead of a treatment dummy indicator in both specifications. The interaction terms are positive and statistically significant in both models, with negative and significant squared terms. This suggests that larger solar system homes sell with higher prices but with diminishing marginal returns, showing a concave relationship between system size and home values.

Table 6: Estimated results using hedonic and repeat-sales models: Solar capacity

VARIABLES	**Dependent Variable: Log of Property Sale Price**					
	Hedonic Model			**Repeat-Sales Model**		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar at time of sale	0.0355	0.0787*	0.0751	0.0549	0.0760	0.0574
	(0.0435)	(0.0466)	(0.0471)	(0.0555)	(0.0533)	(0.0530)
Solar capacity	0.00269***	0.00174*	0.00168*	0.00347	-0.0128	-0.0139
	(0.000485)	(0.00101)	(0.000997)	(0.00458)	(0.0136)	(0.0135)
Solar at time of sale × capacity	0.00975	0.0117	0.0109	-0.00415	-0.00268	-0.00114
	(0.0111)	(0.0110)	(0.0110)	(0.0122)	(0.0116)	(0.0116)
(Solar at time of sale × capacity)^2	-0.000508	-0.000658	-0.000641	0.000286	0.000181	9.71e-05
	(0.000664)	(0.000661)	(0.000655)	(0.000590)	(0.000573)	(0.000567)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,542	271,542	271,542	176,737	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	×	✓	✓	×	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓
Property Controls	✓	✓	✓	✓	✓	✓
Solar Attributes	×	✓	✓	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The treatment variable equals one if a solar system was present at the time of sale. The solar capacity is a continuous variable in kW. Standard errors are clustered at the census block-group level in Models (2)–(3) and at the parcel level in Models (4)–(6). The specification allows the solar premium to vary nonlinearly with system capacity through the interaction between solar-at-sale and capacity and its squared term. The complete table presented in Appendix Table A12

Second, in Appendix Table A5, we replaced continuous capacity with four solar capacity indicator variables. The hedonic and repeat-sales models' estimates are consistent with a monotonic but concave relationship; larger systems are capitalized more, but the incremental premium per additional kW declines as capacity increases. In prior literature, the only study that examined the effect of solar system capacity on home value found no effect (Dastrup et al., 2012). In the Delaware low-penetration context, buyers value the presence of solar more than differences in system size.

5.7 Hedonic and repeat sale model: Metropolitan area

Lastly, we investigate the heterogeneity in solar house price premiums in metropolitan and non-metropolitan housing markets. Table 7 presents estimates from both the simple hedonic and repeat-sales specifications that interact the solar treatment variable with an indicator for metropolitan area (i.e., Wilmington and Dover), while continuing to control for detailed home characteristics and renovation.

Across specifications, the baseline solar effect remains positive and statistically significant, confirming that solar systems are capitalized into housing prices in both metropolitan and non-metropolitan areas. However, the interaction terms reveal meaningful spatial heterogeneity. The repeat-sales Models 4-6 show that the solar home price premium is larger in metropolitan areas, with estimates implying an additional capitalization of approximately 7% compared to non-metropolitan housing markets. These effects are statistically significant at the 10% levels, and robust to the inclusion of home and solar system characteristics, but the effect goes away when we include the renovation controls.

Table 7: Estimated results using hedonic and repeat-sales models: Metropolitan area

VARIABLES	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			Repeat-Sales Model		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar at time of sale	0.100***	0.125***	0.117***	0.0293***	0.0579***	0.0464**
	(0.0125)	(0.0185)	(0.0183)	(0.0113)	(0.0185)	(0.0184)
Metropolitan area	-0.0103	-0.0104	-0.0109	-0.0131	-0.0154	-0.0131
	(0.0185)	(0.0185)	(0.0186)	(0.0291)	(0.0288)	(0.0290)
Solar at time of sale × Metropolitan area	-0.0425*	-0.0278	-0.0265	0.0692*	0.0729*	0.0636
	(0.0228)	(0.0220)	(0.0216)	(0.0420)	(0.0402)	(0.0406)
Constant	10.30***	10.30***	10.29***	11.93***	11.93***	11.95***
	(0.0755)	(0.0755)	(0.0753)	(0.0855)	(0.0855)	(0.0840)
Observations	271,584	271,542	271,542	176,763	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	×	✓	✓	×	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓
Property Controls	✓	✓	✓	✓	✓	✓
Solar Attributes	×	✓	✓	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The treatment variable equals one if a solar system was present at the time of sale. Metropolitan area equals one for properties located within the Dover–Wilmington Metropolitan Statistical Area (MSA) and zero otherwise. Standard errors are clustered at the census block-group level in Models (2)–(3) and at the parcel level in Models (4)–(6). The interaction term allows the solar premium to differ between metropolitan and non-metropolitan housing markets. The complete table presented in Appendix Table A13.

Major U.S. metropolitan housing markets often have higher prices than non-metropolitan areas. There is substantial variation in the economic and geographic conditions of land parcels, not only in proximity to jobs and economic conditions, but also in wide variations in topography, elevation, and proximity to water, open space, and natural amenities (Kok et al., 2014). In dense urban and suburban environments, solar systems may be particularly salient as visible, durable investments that signal both anticipated energy savings and environmental preferences. Moreover, metropolitan buyers are more likely to face higher electricity prices and stronger policy incentives, increasing the present value of future bill savings embedded in solar installations. These mechanisms are consistent with the broader literature showing that capitalization effects of housing attributes vary systematically with local market conditions and spatial context (Kok et al., 2014).

These findings highlight that solar capitalization is not spatially uniform. While solar systems have positive price premiums across Delaware, the magnitude of capitalization is substantially larger in metropolitan markets. This result contributes to the solar hedonic literature by demonstrating that local market structure and spatial context play a critical role in shaping the private financial returns to residential solar adoption, particularly in a low-penetration state such as Delaware.

5.8 Solar capitalization on home sales price

In this section, we discuss the solar capitalization by converting estimated coefficients into dollar-denominated premiums for a solar home. Under a log-linear specification, the percentage capitalization effect is given by pc $= (\exp(\beta_1) - 1) \times 100\%$ (Halvorsen and Palmquist, 1980; Giles, 1982). The average inflation-adjusted solar property sale price and solar system cost are $292,189 and $21,116, normalized to the 2010 HPI and 2010 CPI, respectively. To test solar full capitalization, we hypothesize that the solar home price premium exceeds the average cost of a solar system at the time of home resale. The detailed capitalization calculations are presented in Appendices Table A6, Table A7, and Table A8.

Across the preferred repeat-sales estimates in Table 1, Table 2, & Table 5, we find a solar home premium of around 5.68% to 7.65%, which is equivalent to an additional $16,613 to $22,379 gain at the time of solar home sale for an average 6.5kW solar system. In case of a difference between capitalization of ownership and leasing a solar home, the owned system home shows a higher premium than the leased system home; the owned system home sells for an additional $27,740, and the leased system home sells for $14,765. These results are quite similar to descriptive market narratives that leases impose transfer and contractual restriction at resale, and recent literature notes that leasing systems are discounted relative to owned systems (Begley and Hoen, 2021; Gillingham and Watten, 2024).

Overall, when we compared the solar home premium with the cost of solar system installation, it's economically meaningful. The average cost of a solar system is approximately $21,116 without incentives and $14,781 net of the 30% federal ITC in Delaware. Under the above estimates, solar capitalization recovers roughly 112%-125% of the net-of-incentive system cost and 79%-87% of gross system cost[7], suggesting that solar may be partially or fully capitalized at the time of resale, depending on the model and whether net of incentives or gross system cost is used. This is aligned with our primary hypothesis of full capitalization.

We also convert the premiums into dollars per watt to facilitate comparison to system installation costs per watt. We found that values range from $2 to $4 per watt, which are quite similar to those reported in prior studies (Dastrup et al., 2012; Hoen et al., 2017; Gillingham and Watten, 2024). The average installed solar cost was $3 to $4 per watt on a pre-incentive basis during the sample period in Delaware; these values indicate partial to

[7]The solar investment cost recovery is calculated by taking the ratio of solar premium/ solar system cost. We consider the average solar cost for only the solar treatment group, $21,116 (lower bound), and all solar, $25,672 (upper bound), without the federal ITC incentive (used for gross cost recovery). However, with incentive it ranges from $14,781 to $17,970 (use for net-of-incentive cost recovery).

near-full capitalization in home values.

Finally, we examine heterogeneity in capitalization by solar age and spatial variation across metropolitan and non-metropolitan housing markets. Homes with a solar system 0-3 years old sell for an additional \$13,000 to \$18,000, and systems 4-6 years old sell for \$11,000 to \$13,000, whereas systems older than 7 years show statistically insignificant capitalization. After 4-6 years, solar systems are capitalized at about 72%-82% of the system cost (including federal ITC). However, in metropolitan areas, the solar home price premium ranges from 7.1% to 7.5%, indicating that solar homes sell for an additional \$20,935 to \$22,096 in metropolitan areas compared with non-metropolitan areas. This is comparable to or slightly above the capital recovery rates observed in the full sample. Overall, these suggest that solar is fully capitalized at the time of resale.

6 Conclusion

The rapid growth in residential rooftop solar adoption has created a new class of household energy assets whose benefits extend beyond energy cost savings to include effects on housing markets. To what extent solar systems capitalize on property values remains unclear and unexplored in low-to medium-penetration regions in the US. This remains a central question for households, researchers, policymakers, and solar developers, who have encountered this question asked by prospective solar buyers. Existing solar hedonic studies have documented positive solar premiums in several housing markets in the US. However, the majority of these studies focus on early-adoption, high-penetration regions and are unable to investigate heterogeneity across system attributes in detail, such as age, size, and ownership in capitalization. Moreover, prior work has been largely silent on mechanisms such as technology depreciation, spatial heterogeneity, and the role of renovation controls, which complicate inference when solar adoption is correlated with home improvement behaviors. These gaps motivate our analysis of Delaware, a relatively low-penetration, late-adoption market that offers a valuable test of capitalization under less mature information conditions.

This study uses detailed administrative data on single-family property transactions, historical housing characteristics, renovation permits, and solar installations, and employs hedonic and repeat-sales models to estimate the causal effect of solar installation on home values. We find that homes with a rooftop solar system in Delaware sell at a premium of around 7.6% (\$22,379) when renovation activity is not controlled for. After accounting for renovation, the estimated premium declines to 5.6% (\$16,613), on average, for a 6.5 kW residential solar system. Importantly, the estimated premiums imply substantial cost recovery, with capitalization rates that approximate or exceed net-of-incentive system costs for most installations. We also explore some heterogeneity in solar capitalization. First, the solar system age plays an important role, capitalization declines with age, with nearly full recovery within the first three years of installation and diminishing returns after year six. However, solar value does not disappear with age; rather, it shifts from the capitalized resale value to realized electricity bill savings. Second, the effects of solar system capacity remain less clear, but indicator-based capacity bins reveal nonlinear patterns consistent with diminishing marginal valuation. Third, capitalization rates are higher in metropolitan housing markets, although the effect is weakly statistically significant and becomes statistically insignificant once renovation controls are included. Fourth, owned solar systems generate larger resale premiums than leased systems; however, this difference is weakly statistically significant when we exclude renovation controls. These findings suggest that solar capitalization reflects the remaining economic value of the system, which varies with system age, ownership status, and local housing-market conditions.

We further develop a series of counterfactual policy simulations that combine resale premiums with electricity-bill savings, federal ITC and state GEG incentives, SREC revenues, and electricity export compensation via the net-metering program. We find that policy design significantly affects both the timing and magnitude of homeowner returns. Upfront incentives, particularly the federal ITC and state grants, accelerate investment recovery by lowering the initial cost of solar, while SREC revenues and export compensation increase returns accumulated over the system's operating lifespan. Under the current Delaware policy environment, homeowners recover their investment substantially earlier (at about 10 years) than in the absence of policy support (at about 20 years), although the pre-2026 combination of federal and state incentives produces the largest private returns. In contrast, eliminating major incentives and export compensation substantially delays cost recovery and reduces lifetime net benefits. Together, the simulations show that resale premiums alone do not capture the full financial value of rooftop solar and that both upfront incentives and ongoing benefits play important roles.

This study contributes to the solar hedonic literature in several ways. First, we provide evidence from a low-penetration solar housing market. Second, we introduce a finer-grained dataset for valuing the solar system by incorporating system attributes and home renovation controls, which help address potential omitted-variable bias arising from unobserved home improvements. Third, we observe that the depreciation effect of solar age is economically meaningful. Finally, by combining resale premiums with policy simulations, we move beyond estimating resale premiums alone and evaluate the overall financial returns to rooftop solar under alternative policy scenarios.

The findings have important implications for both policymakers and prospective homeowners. The estimated resale premium represents an additional financial benefit that is rarely incorporated into policy discussions during solar adoption. When combined with electricity bill savings and existing incentive programs, the results suggest that rooftop solar can generate substantial financial returns over the system's lifespan. At the same time, the simulations indicate that removing federal ITC or reducing state incentives substantially delays investment recovery, particularly for households with shorter expected ownership durations. Stable and predictable incentive policies may therefore reduce financial uncertainty and encourage adoption in lower-penetration solar markets where perceived investment risks remain an important barrier.

However, our data have limitations for investigating heterogeneity in capitalization. We do not observe homeowner or buyer demographic characteristics, household-level electricity consumption, financing terms, or leasing contract terms in our data. We also do not observe the home's direction of orientation, the installation direction of solar panels on the roof, or whether the panels are visible from the street. These all highlight important directions for future work. By accessing such rich data, we can explore further how high-income, more educated, ideological, political party, and younger people value the solar system at home. How much are they willing to pay? Are people who have high electricity consumption more likely to pay for a solar home? Do prospective solar home buyers value solar panels' visibility and home appearance?

Overall, the results suggest that residential rooftop solar should be viewed not only as an energy investment but also as a housing asset whose value is jointly determined by electricity markets, housing markets, and public policy. Understanding how these components interact is increasingly important as governments reconsider solar incentives and as residential solar adoption expands into new geographic markets. These findings reduce uncertainty regarding the financial consequences of residential rooftop solar and provide evidence that homeowners can recover a substantial share of their investment through both housing-market capitalization and long-term electricity savings. However, understanding the mechanisms underlying solar capitalization is also relevant to economists, policymakers, and solar market stakeholders, informing future policy implications and motivating homeowners to adopt solar. More broadly, capitalization evidence may help inform the calibration of federal ITC, state-level interconnection rules, appraisal guidelines, and property disclosure regulations.

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A Appendix A

A.1 Literature review

Table A1: Solar home capitalization literature

Study	Geographic Region	Data / Period	Method	Limitation / Future Research
Dastrup et al. (2012)	San Diego and Sacramento, California (USA)	329 solar homes (2003–2010); N=275 (San Diego), N=54 (Sacramento); Permit data with 25 solar homes (2003–2007)	Hedonic and repeat-sales models; Permit analysis uses hedonic model only	No precise solar installation date; Small sample size; Limited renovation permit information
Hoen et al. (2013)	California (USA)	1,894 solar homes (1998–2009)	Hedonic and repeat-sales models	No home renovation controls
Ma et al. (2016)	Perth metropolitan area, Australia	413 solar homes (2009–2012)	Hedonic, repeat-sales, and hybrid models	Examines solar system age effects; No renovation controls
Wee (2016)	Oʻahu, Hawaiʻi (USA)	259 solar homes (2000–2013)	Hedonic and repeat-sales models	No home renovation controls
Hoen et al. (2017)	Eight U.S. states: CA, FL, MD, NC, PA, CT, MA, NY	3,951 solar homes (CA=3,828; FL=25; MD–NC–PA=77; CT–MA–NY=21) (2002–2013)	Hedonic, repeat-sales, and pooled multi-state models	Most observations from California; Limited information for other states; No renovation controls
Qiu et al. (2017)	Phoenix, Arizona (USA)	123 solar homes (2014)	Hedonic and matching methodology	No precise installation date; Limited sample size
Lan et al. (2020)	Southport, Queensland, Australia	316 solar homes (2008–2018)	Hedonic model with matching; Comparison of FiT regimes	–
Begley and Hoen (2021)	California, Connecticut, Massachusetts, Oregon (USA)	45,840 solar homes (2011–2020)	Hedonic and repeat-sales models	–
Best et al. (2021)	Australia (national sample)	Solar homes (2015–2016; 2017–2018)	Hedonic models with financial asset controls	Examines effects of solar system age
Best et al. (2023)	Australia (rental market)	286 solar homes (2015–2016; 2017–2018)	Hedonic models	–
Gillingham and Watten (2024)	Massachusetts and Connecticut (USA)	6,134 solar homes (1990–2019)	Hedonic and panel models; Renovation controls included	Limited renovation controls (new roof, electrical work, kitchen, bathroom, heat pump, geothermal, remodeling, well maintenance)
Asproudis et al. (2024)	United Kingdom	6,230 solar homes (2012–2018)	Hedonic and causal machine learning models	Limited renovation controls (new windows, bathroom, kitchen, roof)

A.2 Additional descriptive statistics

Table A2: Descriptive statistics by solar home status

	Non-Solar Homes				Solar Homes				Mean Diff.	p-value
Variable	Obs.	Mean	SD	Median	Obs.	Mean	SD	Median	Solar–Non-solar	t-test
Sale price	265354	248540.25	145986.98	220000.00	6863	304345.97	158861.63	276489.00	55805.72	0.00
Sale price (Inflation-adjusted 2010 HPI)	265354	240248.86	129582.05	215856.16	6863	288917.00	130226.15	270701.13	48668.15	0.00
Log sale price (Not inflation-adjusted)	265354	12.26	0.59	12.30	6863	12.49	0.54	12.53	0.23	0.00
Log sale price (Inflation-adjusted 2010 HPI)	265354	12.25	0.54	12.28	6863	12.47	0.46	12.51	0.22	0.00
Total number of bathrooms	265354	2.42	1.00	2.00	6863	2.73	0.98	3.00	0.32	0.00
Total number of bedrooms	265354	3.24	0.71	3.00	6863	3.46	0.70	3.00	0.22	0.00
Total land area in acres	265354	0.37	0.79	0.20	6863	0.43	0.82	0.24	0.06	0.00
Log total land area in acres	265354	-1.59	1.01	-1.61	6863	-1.30	0.84	-1.43	0.30	0.00
Total living area, sq. ft.	265159	1985.48	890.22	1775.00	6851	2366.69	1014.76	2152.00	381.21	0.00
Log total living area, sq. ft.	265159	7.50	0.43	7.48	6851	7.68	0.42	7.67	0.18	0.00
Property age	265354	33.08	31.63	24.00	6863	23.56	25.62	16.00	-9.52	0.00
Property age, IHS	265354	3.36	1.70	3.87	6863	2.88	1.77	3.47	-0.47	0.00
Property build year	265354	1978.45	31.47	1988.00	6863	1988.46	25.66	1996.00	10.01	0.00
Pool	265354	0.00	0.06	0.00	6863	0.00	0.06	0.00	0.00	0.77
Total number of stories	264937	1.62	0.52	2.00	6848	1.64	0.50	2.00	0.02	0.00
Property sale year	265354	2011.53	7.89	2012.00	6863	2012.02	7.83	2013.00	0.49	0.00
Solar age					6863	0.71	2.24	0.00	0.71	0.00
Solar capacity, kW					6821	8.12	3.64	7.60	8.12	0.00
Solar system cost					6821	34403.37	14550.49	31459.00	0.00	0.00
Solar system cost (Inflation-adjusted 2010 CPI)					6821	25672.78	15386.77	21795.09	0.00	0.00
Solar installation year					6863	2019.13	4.42	2021.00	0.00	0.00
Solar ownership					6863	0.74	0.43	1.00	0.00	0.00

Note: Solar ownership equals one for owned systems. Robust comparison uses non-solar and solar home groups.

Table A3: Descriptive statistics by solar status at time of sale

	No Solar at Time of Sale				Solar at Time of Sale				Mean Diff.	p-value
Variable	Obs.	Mean	SD	Median	Obs.	Mean	SD	Median	Solar–Non-solar	t-test
Sale price	271371	249532.47	146342.44	220000.00	846	382978.78	163389.53	350000.00	133446.30	0.00
Sale price (Inflation-adjusted 2010 HPI)	271371	241317.78	129808.14	216995.63	846	292182.94	124447.48	270677.69	50865.14	0.00
Log sale price (Not inflation-adjusted)	271371	12.26	0.59	12.30	846	12.77	0.44	12.77	0.50	0.00
Log sale price (Inflation-adjusted 2010 HPI)	271371	12.26	0.54	12.29	846	12.50	0.42	12.51	0.24	0.00
Total number of bathrooms	271371	2.42	1.00	2.00	846	2.79	1.00	3.00	0.37	0.00
Total number of bedrooms	271371	3.25	0.71	3.00	846	3.41	0.74	3.00	0.16	0.00
Total land area in acres	271371	0.37	0.79	0.20	846	0.48	0.90	0.26	0.11	0.00
Log total land area in acres	271371	-1.58	1.01	-1.61	846	-1.21	0.86	-1.35	0.37	0.00
Total living area, sq. ft.	271165	1993.87	894.84	1776.00	845	2383.74	1033.42	2150.00	389.88	0.00
Log total living area, sq. ft.	271165	7.51	0.43	7.48	845	7.69	0.41	7.67	0.18	0.00
Property age	271371	32.84	31.55	24.00	846	30.70	21.96	25.00	-2.15	0.05
Property age, IHS	271371	3.34	1.70	3.87	846	3.81	0.87	3.91	0.47	0.00
Property build year	271371	1978.67	31.40	1988.00	846	1989.52	21.58	1996.00	10.85	0.00
Pool	271371	0.00	0.06	0.00	846	0.01	0.11	0.00	0.01	0.00
Total number of stories	270943	1.62	0.52	2.00	842	1.56	0.54	1.75	-0.06	0.00
Property sale year	271371	2011.51	7.89	2012.00	846	2020.22	3.05	2021.00	8.70	0.00
Solar age	271371	0.00	0.01	0.00	846	5.71	3.49	5.16	5.71	0.00
Solar capacity, kW	271330	0.18	1.34	0.00	845	6.54	2.75	6.24	6.35	0.00
Solar system cost	5932	34530.70	14597.54	31611.00	844	33508.41	14191.72	30773.00	-1022.30	0.06
Solar system cost (Inflation-adjusted 2010 CPI)	5932	26321.17	15737.90	22251.79	844	21115.56	11678.68	18632.93	-5205.61	0.00
Solar installation year	6017	2019.78	4.12	2022.00	846	2014.50	3.64	2015.00	-5.28	0.00
Solar ownership	6017	0.77	0.42	1.00	846	0.51	0.50	1.00	-0.26	0.00

Note: Solar at time of sale indicates that the solar system was installed before or by the property transaction date.

Table A4: Estimated results using hedonic and repeat-sales models: Solar capacity

	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			Repeat-Sales Model		
VARIABLES	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4	(5) Model 5	(6) Model 6
Solar at time of sale × capacity	0.0209***	0.0272***	0.0257***	0.00919**	0.0139***	0.0114**
	(0.00408)	(0.00573)	(0.00560)	(0.00400)	(0.00526)	(0.00519)
(Solar at time of sale × capacity)^2	-0.000957**	-0.00137***	-0.00132***	-0.000394	-0.000664*	-0.000549
	(0.000387)	(0.000467)	(0.000458)	(0.000340)	(0.000377)	(0.000365)
Solar homes (ever)		0.0244***	0.0230***		0.0578	0.0622
		(0.00444)	(0.00438)		(0.0536)	(0.0535)
Solar age		-0.00808***	-0.00707***		-0.00408	-0.00256
		(0.00221)	(0.00218)		(0.00314)	(0.00318)
Total number of bedrooms	0.0210***	0.0209***	0.0205***	-0.00687	-0.00690	-0.0150
	(0.00344)	(0.00344)	(0.00342)	(0.0173)	(0.0173)	(0.0167)
Total number of bathrooms	0.0950***	0.0950***	0.0953***	0.154***	0.154***	0.128***
	(0.00362)	(0.00362)	(0.00361)	(0.0138)	(0.0138)	(0.0134)
Ln(total living area, sq. ft.)	0.276***	0.275***	0.275***	-0.00961	-0.00955	-0.00472
	(0.0113)	(0.0113)	(0.0113)	(0.00702)	(0.00702)	(0.00698)
Property age (IHS)	-0.0603***	-0.0602***	-0.0567***	-0.00342***	-0.00343***	0.00331***
	(0.00264)	(0.00265)	(0.00275)	(0.00113)	(0.00113)	(0.00121)
Property with pool	0.0743***	0.0743***	0.0738***	0.000899	0.000932	0.00119
	(0.0148)	(0.0148)	(0.0148)	(0.0235)	(0.0235)	(0.0234)
Total number of stories	-0.0117	-0.0116	-0.0112	0.0359**	0.0359**	0.0311**
	(0.00718)	(0.00718)	(0.00720)	(0.0155)	(0.0155)	(0.0155)
Ln(total land area, acres)	0.120***	0.120***	0.119***	0.0225**	0.0226**	0.0201*
	(0.00447)	(0.00447)	(0.00447)	(0.0108)	(0.0108)	(0.0109)
Structural work			0.0251***			0.0646***
			(0.00698)			(0.00742)
Bathroom renovation			0.0174			0.119***
			(0.0276)			(0.0417)
Bedroom renovation			0.0370***			0.0883***
			(0.0121)			(0.0211)
Kitchen renovation			0.0465***			0.222***
			(0.00998)			(0.0180)
Major addition			0.0514***			0.0586***
			(0.00565)			(0.00401)
Repair			0.0329*			0.0639**
			(0.0173)			(0.0258)
Roof renovation			0.0297***			0.0894***
			(0.0102)			(0.0134)
Utility work			0.0235***			0.0581***
			(0.00500)			(0.00282)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0845)	(0.0844)	(0.0829)
Observations	271,542	271,542	271,542	176,737	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	×	✓	✓	×	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the logarithm of property sale price. The table reports specifications allowing the solar premium to vary nonlinearly with solar system capacity. Standard errors are clustered at the census block-group level in Models (2)–(3) and at the parcel level in Models (4)–(6).

Table A5: Estimated results using hedonic and repeat-sales models: Solar capacity indicator approach

	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			Repeat-Sales Model		
VARIABLES	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4	(5) Model 5	(6) Model 6
Solar capacity 1–4 kW (N=143)	0.0406**	0.0684***	0.0642***	0.0438	0.0694*	0.0542
	(0.0206)	(0.0246)	(0.0248)	(0.0355)	(0.0381)	(0.0375)
Solar capacity 4–6 kW (N=262)	0.124***	0.153***	0.146***	0.0478*	0.0712**	0.0623**
	(0.0212)	(0.0268)	(0.0269)	(0.0254)	(0.0293)	(0.0295)
Solar capacity 6–9 kW (N=340)	0.0701***	0.0980***	0.0896***	0.0350*	0.0615**	0.0477*
	(0.0156)	(0.0209)	(0.0204)	(0.0194)	(0.0259)	(0.0257)
Solar capacity 9–22 kW (N=130)	0.131***	0.147***	0.137***	0.0527***	0.0737***	0.0601**
	(0.0172)	(0.0199)	(0.0203)	(0.0197)	(0.0235)	(0.0235)
Solar homes (ever)		0.0239***	0.0225***		0.0560	0.0605
		(0.00446)	(0.00439)		(0.0534)	(0.0534)
Solar age		-0.00894***	-0.00788***		-0.00535	-0.00354
		(0.00229)	(0.00228)		(0.00327)	(0.00329)
Total number of bedrooms	0.0210***	0.0210***	0.0205***	-0.00687	-0.00690	-0.0150
	(0.00344)	(0.00344)	(0.00342)	(0.0173)	(0.0173)	(0.0167)
Total number of bathrooms	0.0949***	0.0949***	0.0952***	0.154***	0.154***	0.128***
	(0.00362)	(0.00362)	(0.00361)	(0.0138)	(0.0138)	(0.0134)
Ln(total living area, sq. ft.)	0.276***	0.275***	0.275***	-0.00962	-0.00957	-0.00476
	(0.0113)	(0.0113)	(0.0113)	(0.00702)	(0.00702)	(0.00698)
Property age (IHS)	-0.0603***	-0.0602***	-0.0567***	-0.00341***	-0.00342***	0.00330***
	(0.00264)	(0.00264)	(0.00275)	(0.00113)	(0.00113)	(0.00121)
Property with pool	0.0740***	0.0739***	0.0735***	0.000869	0.000873	0.00113
	(0.0147)	(0.0147)	(0.0147)	(0.0235)	(0.0235)	(0.0234)
Total number of stories	-0.0116	-0.0116	-0.0112	0.0357**	0.0356**	0.0304**
	(0.00717)	(0.00718)	(0.00720)	(0.0155)	(0.0155)	(0.0154)
Ln(total land area, acres)	0.120***	0.120***	0.119***	0.0217**	0.0218**	0.0193*
	(0.00447)	(0.00447)	(0.00447)	(0.0108)	(0.0108)	(0.0109)
Structural work			0.0250***			0.0644***
			(0.00698)			(0.00742)
Bathroom renovation			0.0174			0.119***
			(0.0276)			(0.0417)
Bedroom renovation			0.0371***			0.0884***
			(0.0121)			(0.0211)
Kitchen renovation			0.0464***			0.222***
			(0.00996)			(0.0180)
Major addition			0.0514***			0.0585***
			(0.00566)			(0.00401)
Repair			0.0329*			0.0640**
			(0.0173)			(0.0258)
Roof renovation			0.0298***			0.0896***
			(0.0102)			(0.0134)
Utility work			0.0235***			0.0580***
			(0.00499)			(0.00282)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,584	271,584	271,584	176,763	176,763	176,763
R-squared	0.695	0.696	0.696	0.854	0.854	0.856
Census Block FE	×	✓	✓	×	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the logarithm of property sale price. Solar capacity is categorized into four groups based on installed system size: 1–4 kW, 4–6 kW, 6–9 kW, and 9–22 kW. The omitted category is homes without solar systems. Standard errors are clustered at the census block-group level in Models (2)–(3) and at the parcel level in Models (4)–(6).

Table A6: Solar capitalization on house sale prices in Delaware

Panel A: Capitalization Estimates Based on Table 2						
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Average Solar Property Sale Price ($)	292,189.78	292,189.78	292,189.78	292,189.78	292,189.78	292,189.78
Average Non-Solar Property Sale Price ($)	241,312.25	241,312.25	241,312.25	241,312.25	241,312.25	241,312.25
Solar Coefficient (β)	0.248***	0.125***	0.0715***	0.122***	0.0456***	0.0738***
Percent Premium (%)	28.146	13.315	7.412	12.975	4.666	7.659
Solar Premium ($)	82,239.72	38,904.62	21,656.57	37,912.82	13,632.31	22,379.24
Premium per Watt ($/W)	12.59	5.96	3.32	5.81	2.09	3.43
Panel B: Capitalization Estimates Based on Table 3						
	Model 1	Model 2	Model 3	Model 4	Model 5	
Solar Coefficient (β)	0.206***	0.0875***	0.114***	0.0398***	0.0553***	
Percent Premium (%)	22.875	9.144	12.075	4.060	5.686	
Solar Premium ($)	66,839.35	26,718.50	35,282.54	11,863.67	16,613.22	
Premium per Watt ($/W)	10.24	4.09	5.40	1.82	2.54	
Panel C: Ownership Status Results Based on Table 4						
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
A. Lease Systems						
Coefficient (β)	0.0835***	0.112***	0.106***	0.0290	0.0493**	0.0428*
Percent Premium (%)	8.709	11.851	11.182	2.942	5.054	4.373
Premium ($)	25,445.41	34,628.25	32,673.21	8,597.57	14,765.95	12,777.20
Premium per Watt ($/W)	3.90	5.30	5.00	1.32	2.26	1.96
B. Ownership Premium: Interaction Term						
Coefficient (β)	0.0137	0.0236	0.0198	0.0426*	0.0414*	0.0275
Percent Premium (%)	1.379	2.388	2.000	4.352	4.227	2.788
Premium ($)	4,030.55	6,977.69	5,843.01	12,716.22	12,350.55	8,146.72
Premium per Watt ($/W)	0.62	1.07	0.89	1.95	1.89	1.25
C. Total Ownership Effect: Lease + Ownership Interaction						
Coefficient (β)	0.0972	0.1356	0.1258	0.0716	0.0907	0.0703
Percent Premium (%)	10.208	14.522	13.406	7.423	9.494	7.283
Premium ($)	29,826.96	42,432.88	39,169.60	21,687.95	27,740.64	21,280.18
Premium per Watt ($/W)	4.57	6.50	6.00	3.32	4.25	3.26

Notes: The coefficient β_1 is used to calculate the solar premium as a percentage of the property sale price. In a log-linear model, the percentage premium is calculated as $\%\Delta Price = \left(e^{\beta} - 1\right) \times 100$, following Halvorsen and Palmquist (1980) and Giles (1982). Solar system costs are normalized using the 2010 Consumer Price Index (CPI), and house sale prices are normalized using the 2010 FHFA House Price Index (HPI). We also account for the federal solar investment tax credit, which allows homeowners to claim 30% of the solar installation cost as a credit against their federal tax liability. For all 6,863 solar systems in the sample, the average inflation-adjusted installation cost is $25,672 before incentives and $17,970 after applying the 30% tax credit. For the subset of homes sold with solar systems, the corresponding average installation cost is $21,115 before incentives and $14,780 after incentives. Solar house price premiums are calculated using the average sale price of solar homes ($292,189.78).

Table A7: Solar capitalization on house sale prices in Delaware: Solar age

Estimated Results Based on Table 5						
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Panel A: Solar Age 0–3 Years						
Solar coefficient (β)	0.121***	0.100***	0.0935***	0.0597***	0.0574***	0.0452***
Percent premium (%)	12.862	10.517	9.801	6.152	5.908	4.624
Solar premium for 6.53 kW system ($)	37,582.88	30,729.87	28,637.70	17,974.94	17,262.38	13,510.00
Premium per watt ($/W)	5.76	4.71	4.39	2.75	2.64	2.07
Panel B: Solar Age 4–6 Years						
Solar coefficient (β)	0.0855***	0.0658***	0.0651***	0.0433**	0.0387**	0.0405**
Percent premium (%)	8.926	6.801	6.727	4.425	3.946	4.133
Solar premium for 6.53 kW system ($)	26,081.32	19,872.73	19,654.36	12,929.73	11,529.40	12,076.59
Premium per watt ($/W)	3.99	3.04	3.01	1.98	1.77	1.85
Panel C: Solar Age 7–9 Years						
Solar coefficient (β)	0.0751***	0.0557***	0.0591***	0.0191	0.0149	0.0261
Percent premium (%)	7.799	5.728	6.088	1.928	1.501	2.644
Solar premium for 6.53 kW system ($)	22,788.45	16,736.76	17,788.90	5,634.46	4,386.22	7,726.55
Premium per watt ($/W)	3.49	2.56	2.72	0.86	0.67	1.18
Panel D: Solar Age $\geq$ 10 Years						
Solar coefficient (β)	0.0271	0.00816	0.0113	0.000605	-0.00698	-0.0138
Percent premium (%)	2.747	0.819	1.136	0.061	-0.696	-1.371
Solar premium for 6.53 kW system ($)	8,026.61	2,394.02	3,320.47	176.83	-2,032.38	-4,004.52
Premium per watt ($/W)	1.23	0.37	0.51	0.03	-0.31	-0.61

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The coefficient β is converted to a percentage premium using $\%\Delta Price = \left(e^{\beta} - 1\right) \times 100$. Solar house price premiums are calculated using the average solar-property sale price. Premiums per watt are calculated using the average system size of 6.53 kW.

Table A8: Solar capitalization on house sale prices in Delaware: Metropolitan area

Estimated Results Based on Table 7						
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Panel A: Solar at Time of Sale						
Solar coefficient (β)	0.100***	0.125***	0.117***	0.0293***	0.0579***	0.0464**
Percent premium (%)	10.517	13.315	12.412	2.973	5.961	4.749
Solar premium for 6.53 kW system ($)	30,729.87	38,904.62	36,266.43	8,687.82	17,417.15	13,877.06
Premium per watt ($/W)	4.71	5.96	5.55	1.33	2.67	2.13
Panel B: Metropolitan Area						
Solar coefficient (β)	-0.0103	-0.0104	-0.0109	-0.0131	-0.0154	-0.0131
Percent premium (%)	-1.025	-1.035	-1.084	-1.301	-1.528	-1.301
Solar premium for 6.53 kW system ($)	-2,994.11	-3,023.03	-3,167.57	-3,802.72	-4,465.25	-3,802.72
Premium per watt ($/W)	-0.46	-0.46	-0.49	-0.58	-0.68	-0.58
Panel C: Solar at Time of Sale × Metropolitan Area						
Solar coefficient (β)	-0.0425*	-0.0278	-0.0265	0.0692*	0.0729*	0.0636
Percent premium (%)	-4.161	-2.742	-2.615	7.165	7.562	6.567
Solar premium for 6.53 kW system ($)	-12,157.88	-8,011.01	-7,641.33	20,935.55	22,096.26	19,186.95
Premium per watt ($/W)	-1.86	-1.23	-1.17	3.21	3.38	2.94
Panel D: Effect in Metropolitan Area (Panel A + Panel C)						
Solar coefficient (β)	0.058	0.097	0.091	0.099	0.131	0.110
Percent premium (%)	5.919	10.208	9.472	10.351	13.974	11.628
Solar premium for 6.53 kW system ($)	17,293.33	29,826.96	27,676.66	30,245.85	40,830.54	33,975.26
Premium per watt ($/W)	2.65	4.57	4.24	4.63	6.25	5.20

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The coefficient β is converted to a percentage premium using $\%\Delta Price = \left(e^{\beta} - 1\right) \times 100$. Solar house price premiums are calculated using the average solar-property sale price. Premiums per watt are calculated using the average system size of 6.53 kW. Panel D reports the combined effect for properties in metropolitan areas, calculated as the sum of the baseline solar-at-sale coefficient and the solar-at-sale by metropolitan-area interaction coefficient.

Table A9: Solar system installation costs in Delaware

Measure	Average Cost ($)
Panel A: Nominal Installation Cost	
Solar system cost without incentive	34,403.37
Solar system cost with 30% federal tax credit	24,082.36
Panel B: Inflation-Adjusted Installation Cost (All Solar Homes)	
Solar system cost without incentive	25,672.78
Solar system cost with 30% federal tax credit	17,970.95
Panel C: Inflation-Adjusted Installation Cost (Homes Sold with Solar)	
Solar system cost without incentive	21,115.56
Solar system cost with 30% federal tax credit	14,780.89

Notes: Panel A reports average nominal installation costs. Panel B reports installation costs adjusted to 2010 dollars using the Consumer Price Index (CPI) for all residential solar systems in Delaware. Panel C reports installation costs adjusted to 2010 dollars using the CPI for the subset of homes that were subsequently sold with a solar system installed. The federal solar investment tax credit (ITC) is assumed to equal 30% of the installation cost.

Table A10: Estimated results using hedonic and repeat-sales models by ownership status

	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			**Repeat-Sales Model**		
VARIABLES	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4	(5) Model 5	(6) Model 6
Solar at time of sale	0.0835***	0.112***	0.106***	0.0219	0.0493**	0.0428*
	(0.0122)	(0.0160)	(0.0161)	(0.0147)	(0.0217)	(0.0220)
Solar at time of sale × Own	0.0137	0.0236	0.0198	0.0426*	0.0414*	0.0275
	(0.0199)	(0.0205)	(0.0205)	(0.0244)	(0.0230)	(0.0230)
Solar capacity		0.00180*	0.00171*		-0.0121	-0.0134
		(0.000983)	(0.000967)		(0.0135)	(0.0135)
Solar homes (ever)		0.00908	0.00845		0.154	0.171
		(0.00927)	(0.00913)		(0.148)	(0.146)
Solar age		-0.00979***	-0.00859***		-0.00603*	-0.00411
		(0.00249)	(0.00249)		(0.00328)	(0.00331)
Total number of bedrooms	0.0210***	0.0209***	0.0204***	-0.00682	-0.00687	-0.0150
	(0.00344)	(0.00344)	(0.00342)	(0.0173)	(0.0173)	(0.0167)
Total number of bathrooms	0.0949***	0.0950***	0.0952***	0.154***	0.154***	0.128***
	(0.00362)	(0.00362)	(0.00361)	(0.0138)	(0.0138)	(0.0134)
Ln(total living area, sq. ft.)	0.276***	0.275***	0.275***	-0.00969	-0.00965	-0.00480
	(0.0113)	(0.0113)	(0.0113)	(0.00702)	(0.00702)	(0.00698)
Property age (IHS)	-0.0603***	-0.0602***	-0.0567***	-0.00341***	-0.00343***	0.00330***
	(0.00264)	(0.00264)	(0.00275)	(0.00113)	(0.00113)	(0.00121)
Property with pool	0.0742***	0.0743***	0.0738***	0.00110	0.00106	0.00128
	(0.0147)	(0.0148)	(0.0148)	(0.0234)	(0.0234)	(0.0234)
Total number of stories	-0.0117	-0.0116	-0.0112	0.0355**	0.0358**	0.0310**
	(0.00717)	(0.00718)	(0.00720)	(0.0155)	(0.0155)	(0.0155)
Ln(total land area, acres)	0.120***	0.120***	0.119***	0.0217**	0.0226**	0.0201*
	(0.00447)	(0.00447)	(0.00447)	(0.0108)	(0.0108)	(0.0109)
Structural work			0.0251***			0.0645***
			(0.00697)			(0.00742)
Bathroom renovation			0.0176			0.119***
			(0.0276)			(0.0417)
Bedroom renovation			0.0370***			0.0883***
			(0.0121)			(0.0211)
Kitchen renovation			0.0464***			0.222***
			(0.00998)			(0.0181)
Major addition			0.0514***			0.0586***
			(0.00565)			(0.00401)
Repair			0.0330*			0.0640**
			(0.0173)			(0.0258)
Roof renovation			0.0296***			0.0893***
			(0.0102)			(0.0134)
Utility work			0.0235***			0.0580***
			(0.00500)			(0.00282)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,584	271,542	271,542	176,763	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	✓	✓	✓	×	×	×
Year-Quarter FE	✓	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the logarithm of property sale price. The treatment variable equals one if a solar system was present at the time of sale. The interaction term $Solar_{it} \times Own_{it}$ identifies whether the solar premium differs for owned systems relative to third-party-owned systems. Standard errors are clustered at the census block-group level in the hedonic models and at the parcel level in the repeat-sales models. The hedonic model is: $\log(P_{itb}) = \beta_0 + \beta_1 Solar_{it} + \beta_2 Solar_{it} \times Own_{it} + \gamma' Z_{it} + \eta' X_{it} + \theta' R_{it} + \tau_t + \phi_b + \varepsilon_{itb}$. The repeat-sales model is: $\log(P_{it}) = \beta_0 + \beta_1 Solar_{it} + \beta_2 Solar_{it} \times Own_{it} + \gamma' Z_{it} + \eta' X_{it} + \theta' R_{it} + \tau_t + \phi_i + \varepsilon_{it}$.

Table A11: Estimated results using hedonic and repeat-sales models: Solar age effect

VARIABLES	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			Repeat-Sales Model		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar age 0–3 years (N=335)	0.121***	0.100***	0.0935***	0.0597***	0.0574***	0.0452***
	(0.0154)	(0.0154)	(0.0152)	(0.0155)	(0.0155)	(0.0152)
Solar age 4–6 years (N=278)	0.0855***	0.0658***	0.0651***	0.0433**	0.0387**	0.0405**
	(0.0150)	(0.0153)	(0.0152)	(0.0220)	(0.0194)	(0.0198)
Solar age 7–9 years (N=163)	0.0751***	0.0557***	0.0591***	0.0191	0.0149	0.0261
	(0.0216)	(0.0213)	(0.0213)	(0.0231)	(0.0233)	(0.0238)
Solar age $\geq$ 10 years (N=100)	0.0271	0.00816	0.0113	0.000605	-0.00698	-0.0138
	(0.0199)	(0.0203)	(0.0202)	(0.0389)	(0.0399)	(0.0407)
Solar capacity		0.00185*	0.00175*		-0.0128	-0.0140
		(0.000978)	(0.000962)		(0.0137)	(0.0135)
Solar homes (ever)		0.00863	0.00805		0.164	0.178
		(0.00923)	(0.00908)		(0.150)	(0.147)
Total number of bedrooms	0.0210***	0.0209***	0.0204***	-0.00686	-0.00691	-0.0150
	(0.00344)	(0.00344)	(0.00342)	(0.0173)	(0.0173)	(0.0167)
Total number of bathrooms	0.0949***	0.0950***	0.0953***	0.154***	0.154***	0.128***
	(0.00362)	(0.00362)	(0.00361)	(0.0138)	(0.0138)	(0.0134)
Ln(total living area, sq. ft.)	0.276***	0.275***	0.275***	-0.00969	-0.00961	-0.00477
	(0.0113)	(0.0113)	(0.0113)	(0.00702)	(0.00702)	(0.00698)
Property age (IHS)	-0.0603***	-0.0602***	-0.0567***	-0.00342***	-0.00342***	0.00330***
	(0.00264)	(0.00264)	(0.00275)	(0.00113)	(0.00113)	(0.00121)
Property with pool	0.0741***	0.0742***	0.0738***	0.000880	0.000927	0.00120
	(0.0147)	(0.0148)	(0.0148)	(0.0234)	(0.0234)	(0.0234)
Total number of stories	-0.0117	-0.0116	-0.0112	0.0356**	0.0359**	0.0311**
	(0.00717)	(0.00718)	(0.00720)	(0.0155)	(0.0155)	(0.0155)
Ln(total land area, acres)	0.120***	0.120***	0.119***	0.0218**	0.0226**	0.0201*
	(0.00447)	(0.00447)	(0.00447)	(0.0108)	(0.0108)	(0.0109)
Structural work			0.0251***			0.0647***
			(0.00698)			(0.00742)
Bathroom renovation			0.0173			0.119***
			(0.0276)			(0.0417)
Bedroom renovation			0.0370***			0.0883***
			(0.0121)			(0.0211)
Kitchen renovation			0.0464***			0.222***
			(0.00998)			(0.0180)
Major addition			0.0514***			0.0586***
			(0.00565)			(0.00401)
Repair			0.0330*			0.0640**
			(0.0173)			(0.0258)
Roof renovation			0.0299***			0.0897***
			(0.0102)			(0.0134)
Utility work			0.0235***			0.0580***
			(0.00500)			(0.00282)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,584	271,542	271,542	176,763	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block Group FE	✓	✓	✓	×	×	×
Year-Quarter FE	✓	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the logarithm of property sale price. Solar age groups indicate the age of the solar system at the time of sale. Standard errors are clustered at the census block-group level in Models (1)–(3) and at the parcel level in Models (4)–(6).

Table A12: Estimated results using hedonic and repeat-sales models: Solar capacity

VARIABLES	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			**Repeat-Sales Model**		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar at time of sale	0.0355	0.0787*	0.0751	0.0549	0.0760	0.0574
	(0.0435)	(0.0466)	(0.0471)	(0.0555)	(0.0533)	(0.0530)
Solar capacity	0.00269***	0.00174*	0.00168*	0.00347	-0.0128	-0.0139
	(0.000485)	(0.00101)	(0.000997)	(0.00458)	(0.0136)	(0.0135)
Solar at time of sale × capacity	0.00975	0.0117	0.0109	-0.00415	-0.00268	-0.00114
	(0.0111)	(0.0110)	(0.0110)	(0.0122)	(0.0116)	(0.0116)
(Solar at time of sale × capacity)^2	-0.000508	-0.000658	-0.000641	0.000286	0.000181	9.71e-05
	(0.000664)	(0.000661)	(0.000655)	(0.000590)	(0.000573)	(0.000567)
Solar homes (ever)		0.00953	0.00865		0.163	0.177
		(0.00945)	(0.00931)		(0.149)	(0.147)
Solar age		-0.00932***	-0.00825***		-0.00563*	-0.00382
		(0.00237)	(0.00236)		(0.00332)	(0.00334)
Total number of bedrooms	0.0209***	0.0209***	0.0204***	-0.00688	-0.00691	-0.0150
	(0.00344)	(0.00344)	(0.00342)	(0.0173)	(0.0173)	(0.0167)
Total number of bathrooms	0.0950***	0.0950***	0.0952***	0.154***	0.154***	0.128***
	(0.00362)	(0.00362)	(0.00361)	(0.0138)	(0.0138)	(0.0134)
Ln(total living area, sq. ft.)	0.275***	0.275***	0.275***	-0.00956	-0.00959	-0.00476
	(0.0113)	(0.0113)	(0.0113)	(0.00702)	(0.00702)	(0.00698)
Property age (IHS)	-0.0602***	-0.0602***	-0.0567***	-0.00341***	-0.00342***	0.00331***
	(0.00264)	(0.00264)	(0.00275)	(0.00113)	(0.00113)	(0.00121)
Property with pool	0.0743***	0.0743***	0.0738***	0.000894	0.000899	0.00116
	(0.0148)	(0.0148)	(0.0148)	(0.0235)	(0.0235)	(0.0234)
Total number of stories	-0.0115	-0.0116	-0.0112	0.0359**	0.0359**	0.0311**
	(0.00719)	(0.00719)	(0.00720)	(0.0155)	(0.0155)	(0.0155)
Ln(total land area, acres)	0.120***	0.120***	0.119***	0.0226**	0.0226**	0.0201*
	(0.00447)	(0.00447)	(0.00447)	(0.0108)	(0.0108)	(0.0109)
Structural work			0.0251***			0.0646***
			(0.00698)			(0.00742)
Bathroom renovation			0.0174			0.119***
			(0.0276)			(0.0417)
Bedroom renovation			0.0370***			0.0883***
			(0.0121)			(0.0211)
Kitchen renovation			0.0464***			0.222***
			(0.00998)			(0.0180)
Major addition			0.0514***			0.0586***
			(0.00565)			(0.00401)
Repair			0.0330*			0.0640**
			(0.0173)			(0.0258)
Roof renovation			0.0296***			0.0896***
			(0.0102)			(0.0134)
Utility work			0.0235***			0.0580***
			(0.00500)			(0.00282)
Constant	10.30***	10.30***	10.28***	11.92***	11.92***	11.94***
	(0.0760)	(0.0760)	(0.0757)	(0.0844)	(0.0844)	(0.0829)
Observations	271,542	271,542	271,542	176,737	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	×	✓	✓	×	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the logarithm of property sale price. The treatment variable equals one if a solar system was present at the time of sale. Standard errors are clustered at the census block-group level in Models (2)–(3) and at the parcel level in Models (4)–(6). The specification allows the solar premium to vary nonlinearly with system capacity through the interaction between solar-at-sale and capacity and its squared term.

Table A13: Estimated results using hedonic and repeat-sales models: Metropolitan area

VARIABLES	Dependent Variable: Log of Property Sale Price					
	Hedonic Model			Repeat-Sales Model		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Solar at time of sale	0.100***	0.125***	0.117***	0.0293***	0.0579***	0.0464**
	(0.0125)	(0.0185)	(0.0183)	(0.0113)	(0.0185)	(0.0184)
Metropolitan area	-0.0103	-0.0104	-0.0109	-0.0131	-0.0154	-0.0131
	(0.0185)	(0.0185)	(0.0186)	(0.0291)	(0.0288)	(0.0290)
Solar at time of sale × Metropolitan area	-0.0425*	-0.0278	-0.0265	0.0692*	0.0729*	0.0636
	(0.0228)	(0.0220)	(0.0216)	(0.0420)	(0.0402)	(0.0406)
Solar capacity		0.00179*	0.00169*		-0.0122	-0.0134
		(0.000978)	(0.000961)		(0.0133)	(0.0132)
Solar homes (ever)		0.00918	0.00859		0.157	0.172
		(0.00923)	(0.00907)		(0.144)	(0.143)
Solar age		-0.00869***	-0.00761***		-0.00659*	-0.00466
		(0.00227)	(0.00228)		(0.00342)	(0.00345)
Total number of bedrooms	0.0211***	0.0209***	0.0205***	-0.00687	-0.00692	-0.0150
	(0.00344)	(0.00344)	(0.00342)	(0.0173)	(0.0173)	(0.0167)
Total number of bathrooms	0.0949***	0.0949***	0.0952***	0.154***	0.154***	0.128***
	(0.00362)	(0.00362)	(0.00361)	(0.0138)	(0.0138)	(0.0134)
Ln(total living area, sq. ft.)	0.276***	0.275***	0.275***	-0.00966	-0.00962	-0.00479
	(0.0112)	(0.0113)	(0.0112)	(0.00702)	(0.00702)	(0.00698)
Property age (IHS)	-0.0603***	-0.0602***	-0.0567***	-0.00338***	-0.00339***	0.00333***
	(0.00264)	(0.00264)	(0.00275)	(0.00113)	(0.00113)	(0.00121)
Property with pool	0.0742***	0.0743***	0.0738***	0.000779	0.000726	0.00101
	(0.0147)	(0.0148)	(0.0148)	(0.0234)	(0.0234)	(0.0234)
Total number of stories	-0.0117	-0.0116	-0.0112	0.0357**	0.0359**	0.0311**
	(0.00717)	(0.00718)	(0.00720)	(0.0155)	(0.0155)	(0.0155)
Ln(total land area, acres)	0.120***	0.120***	0.119***	0.0218**	0.0226**	0.0201*
	(0.00448)	(0.00448)	(0.00448)	(0.0108)	(0.0108)	(0.0109)
Structural work			0.0251***			0.0646***
			(0.00698)			(0.00742)
Bathroom renovation			0.0178			0.119***
			(0.0276)			(0.0417)
Bedroom renovation			0.0371***			0.0880***
			(0.0121)			(0.0211)
Kitchen renovation			0.0464***			0.222***
			(0.00997)			(0.0181)
Major addition			0.0514***			0.0586***
			(0.00565)			(0.00401)
Repair			0.0332*			0.0640**
			(0.0173)			(0.0258)
Roof renovation			0.0294***			0.0899***
			(0.0102)			(0.0134)
Utility work			0.0236***			0.0580***
			(0.00500)			(0.00282)
Constant	10.30***	10.30***	10.29***	11.93***	11.93***	11.95***
	(0.0755)	(0.0755)	(0.0753)	(0.0855)	(0.0855)	(0.0840)
Observations	271,584	271,542	271,542	176,763	176,737	176,737
R-squared	0.695	0.695	0.696	0.854	0.854	0.856
Census Block FE	×	✓	✓	×	×	×
Year-Quarter FE	×	✓	✓	✓	✓	✓
Property FE	×	×	×	✓	✓	✓
Renovation Controls	×	×	✓	×	×	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The dependent variable is the logarithm of property sale price. The treatment variable equals one if a solar system was present at the time of sale. Metropolitan area equals one for properties located within the Philadelphia–Camden–Wilmington Metropolitan Statistical Area (MSA) and zero otherwise. Standard errors are clustered at the census block-group level in Models (2)–(3) and at the parcel level in Models (4)–(6). The interaction term allows the solar premium to differ between metropolitan and non-metropolitan housing markets.

Table A14: Robustness check: hedonic and repeat-sales estimates restricting property transactions to 2009–2025

	Dependent Variable: Log of Property Sale Price				
VARIABLES	Hedonic Model			Repeat-Sales Model	
	Model 1	Model 2	Model 3	Model 4	Model 5
Solar at time of sale	0.213***	0.0854***	0.109***	0.0162	**0.0670*** **
	(0.0184)	(0.00981)	(0.0168)	(0.0170)	(0.0240)
Solar capacity			0.00326***		-0.00757
			(0.00114)		(0.0131)
Solar homes (ever)			-0.000854		0.133
			(0.0113)		(0.149)
Solar age			-0.00784***		-0.0140***
			(0.00215)		(0.00446)
Total number of bedrooms		0.0205***	0.0203***	-0.00377	-0.00382
		(0.00354)	(0.00354)	(0.0203)	(0.0203)
Total number of bathrooms		0.105***	0.105***	0.148***	0.148***
		(0.00348)	(0.00348)	(0.0169)	(0.0169)
Ln(total living area, sq. ft.)		0.254***	0.254***	-0.0296***	-0.0296***
		(0.00985)	(0.00985)	(0.0100)	(0.0100)
Property age (IHS)		-0.0700***	-0.0699***	-0.0248***	-0.0248***
		(0.00316)	(0.00316)	(0.00254)	(0.00254)
Property with pool		0.0779***	0.0779***	0.0232	0.0229
		(0.0145)	(0.0145)	(0.0362)	(0.0362)
Total number of stories		-0.0162**	-0.0161**	0.0401*	0.0414*
		(0.00740)	(0.00742)	(0.0242)	(0.0242)
Ln(total land area, acres)		0.119***	0.118***	0.0113	0.0130
		(0.00476)	(0.00476)	(0.0157)	(0.0156)
Structural work	0.263***	0.0213***	0.0215***	0.0771***	0.0774***
	(0.00868)	(0.00737)	(0.00741)	(0.0117)	(0.0117)
Bathroom renovation	-0.0111	0.0327	0.0332	0.119**	0.120**
	(0.0479)	(0.0291)	(0.0290)	(0.0505)	(0.0504)
Bedroom renovation	0.0555**	0.0508***	0.0506***	0.102***	0.101***
	(0.0231)	(0.0140)	(0.0140)	(0.0284)	(0.0284)
Kitchen renovation	-0.0402**	0.0604***	0.0604***	0.296***	0.297***
	(0.0181)	(0.0113)	(0.0113)	(0.0234)	(0.0235)
Major addition	0.337***	0.0354***	0.0353***	0.0724***	0.0726***
	(0.00474)	(0.00629)	(0.00630)	(0.00699)	(0.00700)
Repair	0.181***	0.0552***	0.0551***	0.0786	0.0777
	(0.0380)	(0.0210)	(0.0210)	(0.0548)	(0.0548)
Roof renovation	0.0997***	0.0377***	0.0352***	0.116***	0.115***
	(0.0157)	(0.0107)	(0.0107)	(0.0195)	(0.0195)
Utility work	0.193***	0.0260***	0.0256***	0.0993***	0.0995***
	(0.00381)	(0.00594)	(0.00595)	(0.00522)	(0.00522)
Constant	12.19***	10.46***	10.46***	12.08***	12.08***
	(0.00151)	(0.0688)	(0.0688)	(0.110)	(0.110)
Observations	163,324	162,923	162,889	78,958	78,936
R-squared	0.058	0.674	0.674	0.829	0.829
Census Block FE	×	✓	✓	×	×
Year-Quarter FE	×	✓	✓	✓	✓
Property FE	×	×	×	✓	✓

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

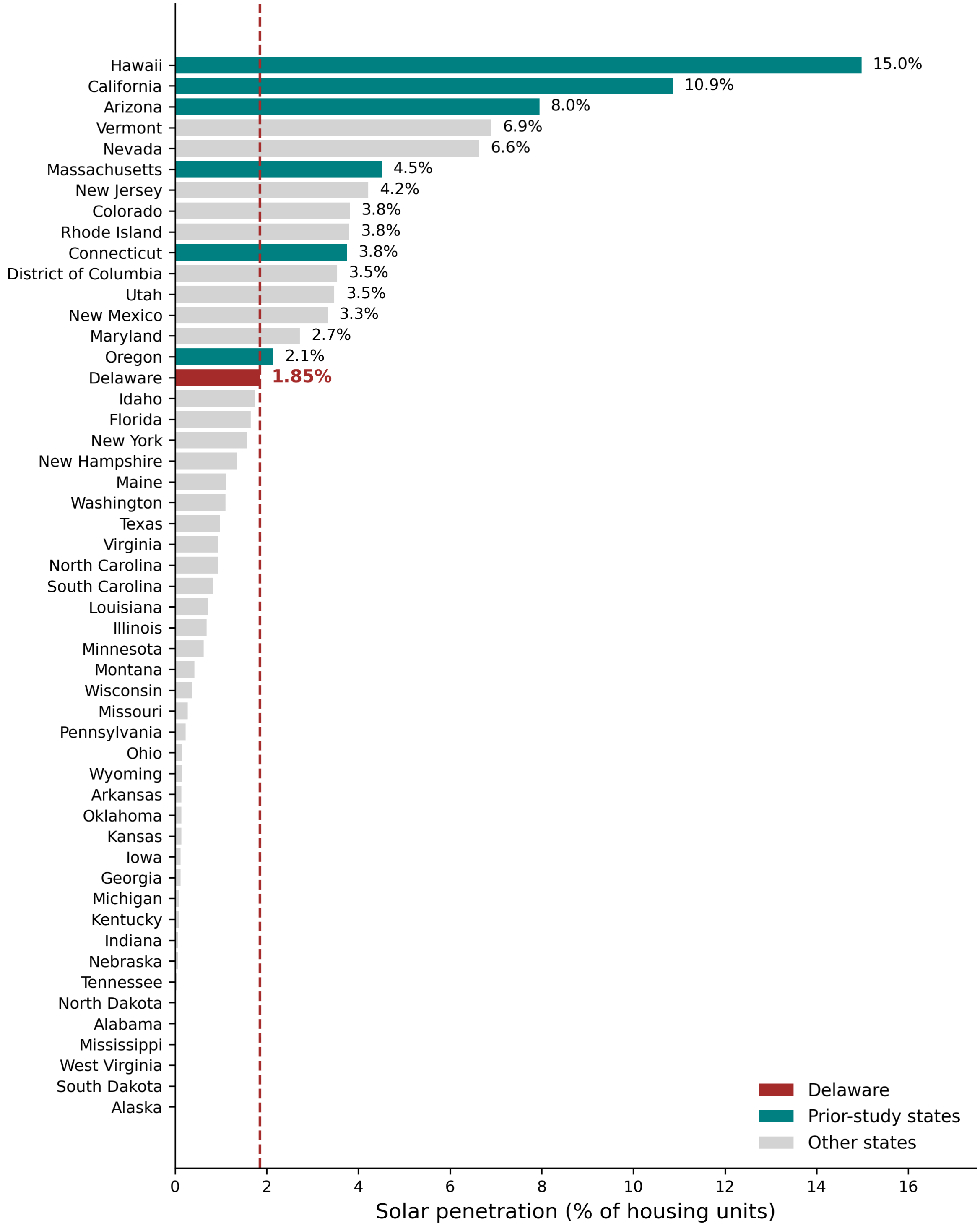

Figure A1: Residential rooftop solar penetration by state

Figure A1 reports residential rooftop solar penetration rates across U.S. states. Prior studies of solar capitalization have primarily focused on states with relatively mature residential solar markets, including Hawaii (15.0%), California (10.9%), Arizona (8.0%), Massachusetts (4.5%), and Connecticut (3.8%). By comparison, Delaware's penetration rate is only 1.85%, substantially below that of these commonly studied markets. Despite its lower adoption rate, Delaware has more than 8,792 residential solar installations, providing sufficient variation to estimate capitalization effects while allowing examination of housing market responses in a less mature solar market.

Table A15: Renovation activity within five years prior to sale, by solar adoption status

Renovation Category	**Full Sample** (%)	**Non-Solar** (%)	**Solar** (%)	**Difference** (pp)
Structural Work	1.77	1.73	3.32	1.59
Bathroom Renovation	0.05	0.05	0.09	0.04
Bedroom Renovation	0.25	0.24	0.41	0.17
Demolition	0.43	0.42	0.68	0.26
Energy-Related Work	0.02	0.02	0.23	0.21
Kitchen Renovation	0.39	0.39	0.61	0.22
Major Addition	6.67	6.58	10.21	3.63
Repair Work	0.09	0.09	0.16	0.07
Roof Renovation	0.50	0.46	2.13	1.67
Utility Work	13.52	13.13	28.60	15.47
Other Improvements	12.93	12.67	23.04	10.37
Observations		272,217		
Solar Homes		6,863		
Non-Solar Homes		265,354		

Notes: The table reports the percentage of properties with renovation permits issued within five years prior to sale. Solar homes are properties that had adopted rooftop solar at some point during the study period. The final column reports the difference in renovation prevalence between solar and non-solar homes.

Table A15 and Table A16 provide descriptive evidence on the relationship between rooftop solar adoption and renovation activity. Across nearly all renovation categories, solar homes exhibit a higher prevalence of recent renovation permits than non-solar homes. The largest differences are observed for utility work, other improvements, major additions, roof renovations, and structural work. For example, 28.6% of solar homes underwent utility-related improvements within five years prior to sale, compared with only 13.1% of non-solar homes. Similarly, major additions were observed in 10.2% of solar homes but only 6.6% of non-solar homes, while roof renovations occurred in 2.1% of solar homes compared with 0.5% of non-solar homes.

Table A16 further highlights the magnitude of these differences by reporting relative likelihood ratios. Solar homes are approximately 4.6 times more likely to have undergone roof renovations, 2.2 times more likely to have experienced utility-related improvements, and nearly twice as likely to have received structural improvements than non-solar homes. These patterns are consistent with the hypothesis that solar adoption often occurs alongside broader reinvestment in housing quality rather than as an isolated improvement. Because many renovation activities—including roof replacement, structural work, and major additions—are themselves associated with higher housing values, failure to account for renovation activity may attribute part of the value of these concurrent investments to rooftop solar systems. Consequently, these descriptive findings provide additional motivation for incorporating renovation permit controls into the hedonic and repeat-sales models presented in the main text.

Table A16: Relative likelihood of renovation activity among solar homes

Renovation Category	**Solar Homes** (%)	**Non-Solar Homes** (%)	**Ratio** (Solar/Non-Solar)
Roof Renovation	2.13	0.46	4.63
Structural Work	3.32	1.73	1.92
Major Addition	10.21	6.58	1.55
Utility Work	28.60	13.13	2.18
Energy-Related Work	0.23	0.02	11.50
Other Improvements	23.04	12.67	1.82

Notes: The ratio column reports the prevalence of renovation activity among solar homes relative to non-solar homes. For example, solar homes are approximately 4.6 times more likely to have undergone roof renovations and 2.2 times more likely to have experienced utility-related improvements. These patterns suggest that solar adoption often occurs alongside broader reinvestment in housing quality.

A.3 Repeat sale model: placebo test

In this section, we conduct a series of placebo tests to check whether the above estimates capture a true capitalization effect rather than correlations. These tests help us to verify that our empirical strategy would falsely detect a solar premium when solar status is randomly assigned.

First, we randomly assign the solar-home treatment indicator "solar at sale" to approximately 800 non-solar homes across Delaware while maintaining the adoption rate, and we employ a repeat-sales model that includes property characteristics. Table A17, Models 1 and 2, show that the placebo solar coefficients are not statistically significant, confirming that the hedonic model does not exhibit solar capitalization when treatment is randomly assigned, even after controlling for solar and property attributes. Second, we explore the temporal dimension of adoption. We construct a solar-home treatment indicator by randomly assigning solar to non-solar homes in a way that matches the distribution of the number of real solar installations by year. The results are presented in Models 3 and 4, which also show that the "placebo solar year" variable has no statistically significant association between solar and sale prices, suggesting that assigning solar by year does not drive our main results. Third, we develop a more balanced solar-home treatment indicator by constructing artificial measures of solar capacity, solar age, and solar ownership-by assigning them to non-solar homes in a way that mirrors the joint distribution of these characteristics among actual solar homes. Models 5 and 6, the "Placebo solar clone" coefficient estimate also indicates no solar effect on property values. Overall, these placebo tests provide strong evidence that the estimated 5% to 7% premium in the main specifications reflects a causal effect of solar installation on home sale price.

Table A17: Placebo tests using repeat-sales models

VARIABLES	**Dependent Variable: Log of Property Sale Price**					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Placebo solar	-0.0125	-0.0125				
	(0.0129)	(0.0129)				
Placebo solar year			-0.00434	-0.00433		
			(0.0168)	(0.0168)		
Placebo solar clone					-0.0276	-0.00873
					(0.0261)	(0.0142)
Constant	11.90***	11.90***	11.90***	11.90***	11.90***	11.90***
	(0.0838)	(0.0838)	(0.0841)	(0.0841)	(0.0838)	(0.0838)
Observations	176,763	176,737	175,898	175,873	176,763	176,763
R-squared	0.854	0.854	0.854	0.854	0.854	0.854
Property FE	✓	✓	✓	✓	✓	✓
Year-Quarter FE	✓	✓	✓	✓	✓	✓
Property Controls	✓	✓	✓	✓	✓	✓
Solar Attributes	×	✓	×	✓	✓	×
Own Interaction	×	×	×	×	✓	×

Notes: Robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The treatment variables are placebo solar indicators constructed to test whether artificial solar assignment predicts changes in home sale prices. Standard errors are clustered at the parcel level in all models.

A.4 Federal and state incentives for residential rooftop solar

Federal and state policies have played an important role in reducing the private cost of residential rooftop solar and increasing the financial returns available to homeowners. At the federal level, the Energy Policy Act of 2005 established a 30% residential solar federal ITC under Section 25D of the Internal Revenue Code, initially subject to a $2,000 cap and effective beginning in 2006. The Energy Improvement and Extension Act of 2008 subsequently extended the credit and removed the $2,000 cap for systems placed in service beginning in 2009. Later federal legislation, including legislation enacted in 2020, extended the credit, and the Inflation Reduction Act (IRA) of 2022 restored the residential clean-energy credit to 30% for qualifying expenditures. Under the rules applicable to systems installed from 2022 through 2025, homeowners could claim a credit equal to 30% of qualified solar installation expenditures, with no general dollar cap for residential solar. For example, depending on installation cost, a 3–20 kW residential system could save several thousand dollars ($3,330 to $19,500) in federal tax benefits. The One Big Beautiful Bill Act of 2025 accelerated the termination of the Section 25D Residential Clean Energy Credit, and the credit is no longer available for expenditures made after December 31, 2025.

Delaware provides several additional forms of support for residential solar, including SRECs, Green Energy Program grants, targeted low- to moderate-income solar assistance, and net metering. Delaware's Renewable Energy Portfolio Standards Act was adopted in 2005 and most recently amended by Senate Bill 265, signed on September 5, 2024. The Act requires utilities to obtain an increasing share of electricity from renewable resources and permits compliance through the acquisition of Renewable Energy Credits (RECs) and SRECs. Delaware's current Renewable Portfolio Standard requires 40% renewable electricity by 2035, including a 10% solar requirement.

The SREC program provides an additional generation-based source of revenue to solar owners. One SREC is created for each megawatt-hour (1,000 kWh) of electricity produced by an eligible solar system, independently of whether the electricity is consumed on site or exported to the grid. A typical residential system generates approximately 7–12 SRECs annually. The SREC Delaware Program was established in 2012 and purchases SRECs through a competitive procurement process, primarily for Delmarva Power to satisfy its Renewable Portfolio Standard obligations. Successful participants enter contracts under which SRECs are transferred in exchange for payment. SREC prices therefore depend on the applicable procurement round and contract structure rather than representing a fixed state subsidy.

A second major state incentive is Delaware's Green Energy Program grants (GEG), which began in 1999 following electricity-sector restructuring and the establishment of the Green Energy Fund. The program provides rebates that offset the installed cost of renewable-energy technologies, including residential solar systems. The Delmarva Power portion of the program is funded through a public-benefits charge collected from customers, while the Delaware Electric Cooperative and participating municipal utilities operate related programs. Residential solar incentive levels have changed over time with program rules and market conditions. Under the grant schedules relevant to our analysis, incentives were capacity based and could offset several thousand dollars of the cost of a residential installation; for example, an incentive of \$0.80/W would correspond to approximately \$2,400 for a 3 kW system and would reach a \$10,000 cap for sufficiently large eligible systems[8]. Current incentive schedules differ from these historical levels; for example, the current Delmarva Power residential PV incentive is \$0.70/W with a maximum award of \$6,000. An important feature of the current Green Energy Program is that grant applicants assign their SRECs to the Delaware Sustainable Energy Utility, meaning that a household generally cannot simultaneously receive the grant and independently retain the same SREC revenue stream.

Delaware also operates a LMI Solar Program intended to extend solar access to households that have historically been underserved by existing renewable-energy programs. DNREC launched the LMI Solar Pilot Program on July 1, 2022. Low-income households qualify through the Weatherization Assistance Program and may receive a cost-free solar installation of up to 4 kW, together with eligible weatherization services. Moderate-income households may receive systems of up to 6 kW, with the LMI Solar Program covering 70% of the installation cost and the homeowner responsible for the remaining 30%. Low-income eligibility is based on income at or below 200% of the federal poverty level, while moderate-income eligibility is based on county-specific state area median income thresholds. For households of one to eight persons, current low-income thresholds range from \$29,160 to \$101,120. Moderate-income thresholds range from \$64,250 to \$121,150 in New Castle County, \$50,300 to \$94,850 in Kent County, and \$54,750 to \$103,200 in Sussex County.

Finally, Delaware's net-metering policy determines the value households receive for electricity generated in excess of contemporaneous household consumption. Under net metering, electricity generated by rooftop solar first offsets electricity that would otherwise be purchased from the utility; excess generation is exported to the distribution grid and credited against subsequent electricity consumption. Delaware law provides for excess kilowatt-hour credits to carry forward to later billing periods. For commission-regulated utilities, beginning January 1, 2024, excess generation is valued using the volumetric components of electricity supply and distribution charges, excluding charges associated with societal-benefit programs and fixed monthly customer charges. Net metering therefore provides an additional source of economic value beyond direct self-consumption and operates separately from the SREC market. For context, residential electricity prices in Delaware averaged approximately 17.45 cents/kWh in January 2026 and have varied over time and across utilities and rate schedules.

[8]Green Energy Program incentive schedules have changed repeatedly over the study period. The historical \$0.80/W and \$10,000 maximum should therefore be associated with the specific program year or incentive schedule used in the empirical simulation. Current Delmarva Power residential incentives differ from these historical values.

A.5 Net private benefits under alternative policy scenarios (extend)

A.5.1 Alternative federal and state solar incentive scenarios

Figure A2 shows the role of federal and state incentives by varying the federal ITC, Delaware grant support, and treatment of SREC revenues while holding simulation assumptions constant. The results demonstrate that these instruments can affect homeowner returns through different channels. Upfront incentives, such as the federal ITC and state grants, reduce the effective installation cost immediately, while SREC revenues provide a stream of benefits as the system generates electricity over its operating lifespan. The historical GEG high-grant ($10,557.78) scenario produces the largest net private benefits throughout the simulated period, reducing the initial investment burden. In contrast, removing the federal ITC produces the lowest net benefits among the incentive scenarios.

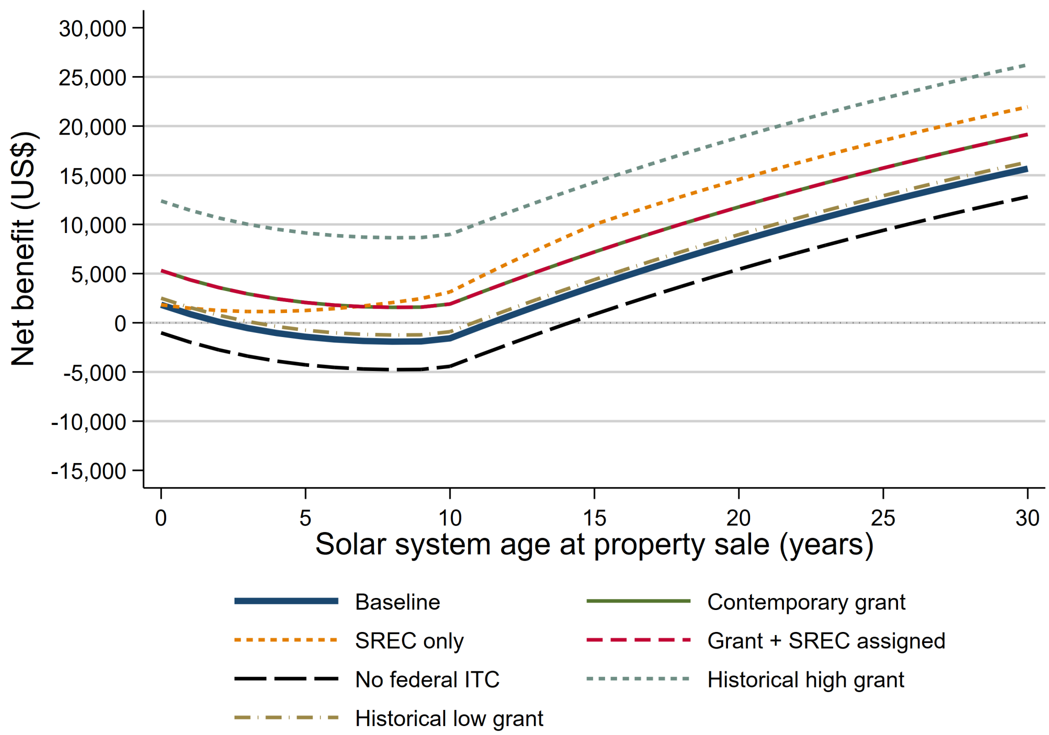

Figure A2: Net private benefits under alternative federal and state solar incentive scenarios

Notes: The scenarios vary federal ITC, Delaware GEG ($3,482.98), and SREC treatment while holding system characteristics and other simulation assumptions constant. (1) Baseline represents the benchmark policy assumptions, including savings on the electricity bill, resale premium, ITC, net metering, but no GEG, and no SREC. (2) Contemporary grant applies baseline and the contemporary Delaware GEG schedule. (3) SREC only removes the GEG grant while allowing the homeowner to retain SREC revenues. (4) Grant + SREC assigned provides the applicable grant but assigns the associated SREC rights rather than allowing the homeowner to retain future SREC revenues. (5) No federal ITC removes the 30% federal ITC and SREC revenues while retaining the other baseline assumptions. (6) Historical high grant ($10,557.78) and historical low grant ($670.08) apply alternative historical Delaware grant levels to illustrate the effect of changes in upfront state support.

A low grant ($670.08) provides substantially less improvement relative to the baseline. Because a grant is received at or near installation, its effect is apparent from the beginning of system ownership and persists throughout the simulated holding period. These counterfactuals show that changes in grant generosity can alter homeowners' financial position, particularly in the early years after installation. SREC revenues accrue with solar generation rather than exclusively at installation and therefore contribute to cumulative returns over time. The distinction between retaining SREC rights and assigning those rights in connection with a grant is therefore important: an upfront grant can improve the homeowner's initial financial position, while assigning future SREC rights reduces the subsequent stream of operating revenues. Thus, grant and SREC policies should not necessarily be interpreted as additive benefits when program participation requires the transfer of SREC rights.

A.5.2 Electricity export-compensation counterfactuals

Figure A3 shows how alternative compensation mechanisms for excess solar electricity affect homeowner returns. Export-compensation policies affect the value generated during system operation. The simulations distinguish between policies that reduce the cost of adopting solar and policies that determine the value of electricity produced after installation.

The results show that alternative export-compensation arrangements generate relatively similar net-benefit paths compared with the much larger effect of removing the broader package of solar incentives. Under retail, partial-retail, avoided-cost, zero-export, and feed-in-tariff compensation, net benefits generally follow a similar life-cycle pattern: they decline in the early years as the estimated resale premium depreciates and then increase as cumulative electricity savings and other operating benefits become more important.

An important reducing export compensation does not eliminate the economic value of electricity the household consumes directly. Solar generation that offsets contemporaneous household consumption still avoids electricity purchases at the retail rate. Export-compensation policy applies only to generation exceeding household consumption during the relevant period. Even zero compensation for exported electricity does not eliminate electricity-bill savings from self-consumed solar generation.

The feed-in-tariff and alternative export-compensation scenarios further shows that the economic importance of export policy depends on the amount of solar generation exported rather than consumed on site. The no-incentives/export-credit counterfactual differs qualitatively. Removing the broader set of policy benefits produces substantially lower net private benefits and delays the break-even point to approximately year 20. Nevertheless, net benefits eventually become positive in the simulation because homeowners continue to realize value from self-consumed electricity and the housing-market resale premium. This result help to distinction between the underlying private value generated by the solar system and the additional value created or accelerated by policy support.

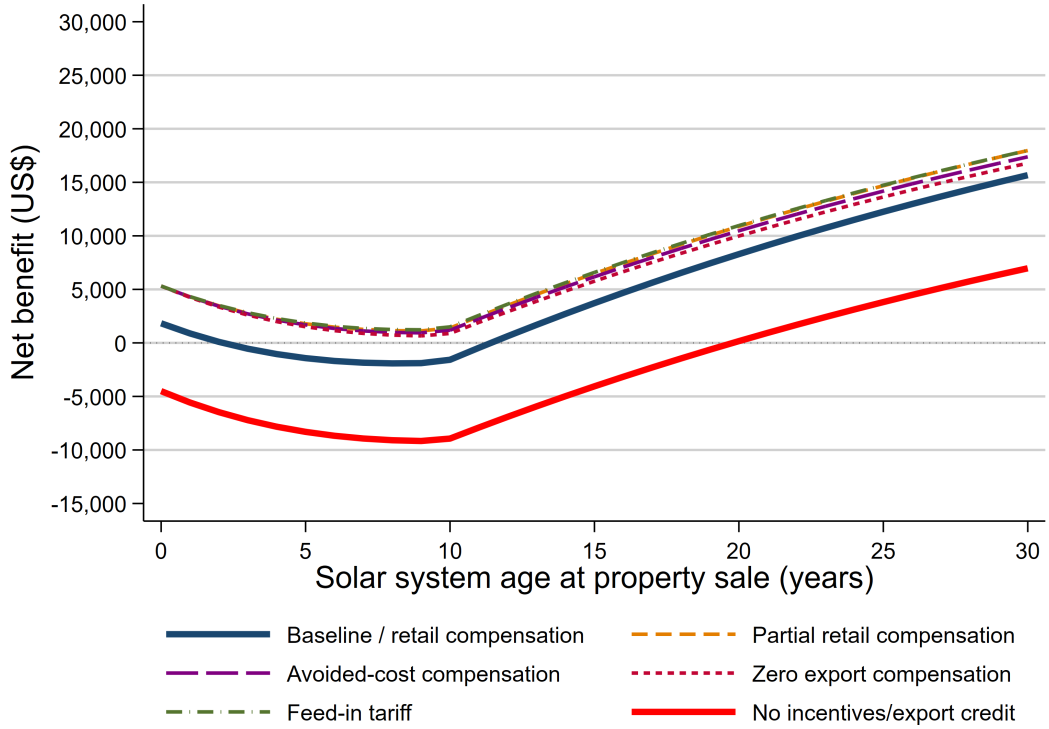

Figure A3: Net private benefits under alternative electricity export-compensation scenarios

Notes: The scenarios vary the compensation received for exported electricity while holding the representative system, household electricity demand, resale-premium estimates, and other simulation assumptions constant. Baseline/retail compensation applies the benchmark retail-rate export-credit assumption. Partial retail compensation compensates exported electricity at a fraction of the retail electricity rate. Avoided-cost compensation values exports at the assumed utility avoided-cost rate rather than the full retail rate. Zero export compensation assigns no monetary credit to electricity exported to the grid. Feed-in tariff compensates exported generation at the assumed feed-in-tariff rate. No incentives/export credit removes the major modeled policy incentives and compensation for exported electricity, representing the lowest-support counterfactual. All parameters not explicitly modified under a given scenario are held at their baseline values.

A.6 Housing turnover among solar and non-solar homes

In Figure A4, we compare annual housing turnover rates for solar and non-solar residential properties in Delaware. Turnover is defined as the number of homes sold in a given year relative to the corresponding housing stock. Solar homes show substantially lower turnover than non-solar homes throughout the period shown, although the difference narrows considerably over time. Rooftop solar adoption expanded over 2009-2025, meaning that the composition and age distribution of solar properties also changed. In the earlier years, many solar properties had only recently installed their systems and therefore had relatively little time to enter the resale market. As the installed solar stock matured, more solar properties became available for resale. The increasing turnover rate among solar homes is consistent with the maturation of Delaware's residential solar market.

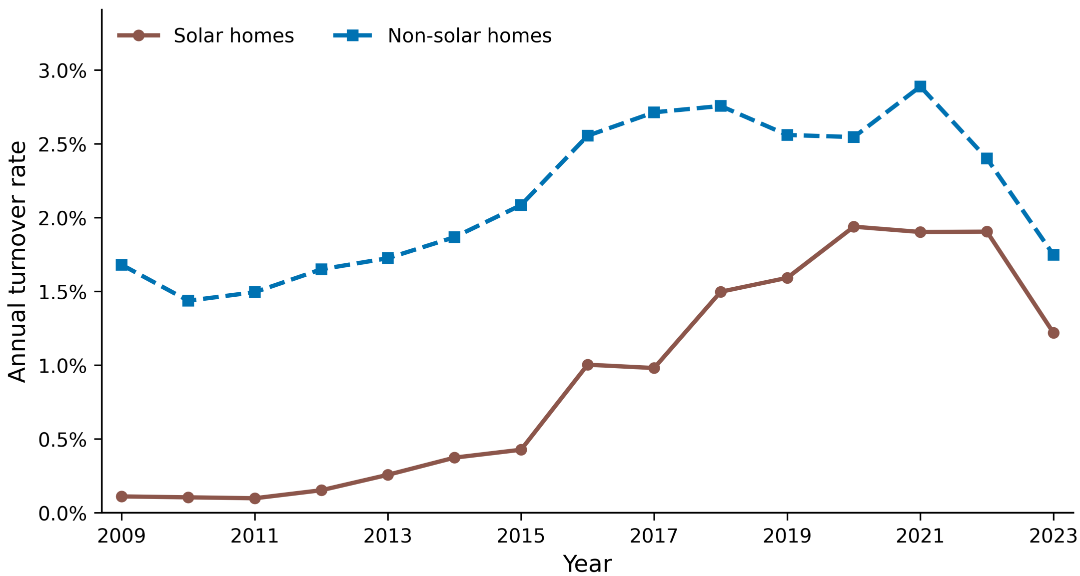

Figure A4: Annual housing turnover rates for solar and non-solar homes in Delaware

A.7 Growth and maturation of Delaware's solar housing stock

In Figure A5, shows the solar-home turnover pattern in the context of the rapid expansion of Delaware's residential solar market. The installed solar housing stock increases from 2,706 properties in 2009 to 6,728 properties in 2023, an increase of approximately 149%. At the same time, the annual turnover rate among solar properties rises from approximately 0.1% in 2009 to around 1.9% during 2020–2022. The simultaneous growth in the installed stock and turnover rate is important for interpreting the solar resale sample.

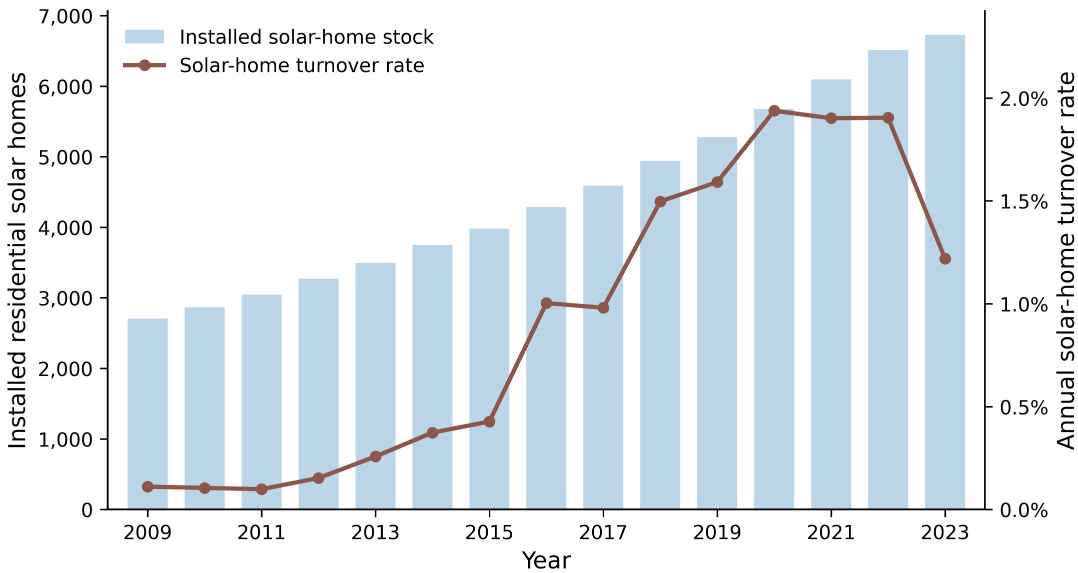

Figure A5: Growth of the residential solar housing stock and annual turnover in Delaware